Working Notes on A_∞ Chiral Algebras and 3D Holomorphic-Topological QFT

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Volume I took a 1-form and followed where it led: into the bar complex, the five theorems, the shadow tower. These notes are about what happens when the 1-form acquires a second direction, what the bar complex becomes when read in three dimensions, and why the coproduct contains more physics than the differential.

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1 Two directions

The bar complex of a chiral algebra $\mathcal{A}$ on a curve X carries two structures.

The first is the **differential** $d_{\overline{B}}$. It extracts OPE residues at collision divisors in $\mathrm{FM}_k(X)$. Points approach each other along the curve. The propagator $\eta_{ij} = d\log(z_i - z_j)$ detects each collision and records its residue. Arnold's relation kills triple residues, so $d_{\overline{B}}^2 = 0$ at genus 0. This is the holomorphic direction. It is everything Volume I studied.

The second is the **coproduct** Δ. On the tensor coalgebra $T^c(s^{-1}\bar{\mathcal{A}})$, deconcatenation is

$$\Delta[a_1 \mid \cdots \mid a_n] \;=\; \sum_{i=0}^{n} [a_1 \mid \cdots \mid a_i] \;\otimes\; [a_{i+1} \mid \cdots \mid a_n].$$

This is the ordered splitting of a word into two subwords. It does not depend on the curve, the OPE, or the central charge. It depends on the ordering. Points are separated along a line, and the line is cut at one of the $n+1$ positions between them. This is the topological direction.

The differential lives on $\mathrm{FM}_k(\mathbb{C})$. The coproduct lives on $\mathrm{Conf}_k(\mathbb{R})$. Together: a bar element of degree k is parametrised by

$$\mathrm{FM}_k(\mathbb{C}) \;\times\; \mathrm{Conf}_k(\mathbb{R}),$$

the product of holomorphic and topological configuration spaces.

*

This product is the operadic fingerprint of a 3d holomorphic-topological QFT on $\mathbb{C}_z \times \mathbb{R}_t$. Observables factorise holomorphically in z and associatively in t. The two-coloured Swiss-cheese operad $\mathsf{SC}^{\mathrm{ch,top}}$ has operation spaces $\mathrm{FM}_k(\mathbb{C}) \times \mathsf{E}_1(m)$: the closed colour is the Fulton–MacPherson compactification (holomorphic collisions); the open colour is the little intervals operad (topological splittings). The bar differential is the closed colour. The bar coproduct is the open colour.

The same 1-form η_{ij} generates both structures. Its collision residues produce the differential. Its ordering (which point comes first along the time axis) produces the coproduct. One form, two directions, one product.

*

At genus 0, $d_{\overline{B}}^2 = 0$ and Δ is coassociative and compatible with $d_{\overline{B}}$: these are the Swiss-cheese relations without obstruction. At genus $g \geq 1$, the curvature $d_{\mathrm{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g$ from Volume I infects the product:

$$m_1^2(a) \;=\; [m_0, a], \qquad m_0 = \kappa(\mathcal{A}) \cdot \omega_g.$$

The Swiss-cheese algebra becomes *curved*. The failure of formality at genus $g \geq 1$ is the bar construction's single most important feature. It is not a defect. It is κ.

2 The recognition theorem

A factorisation algebra on $\mathbb{C} \times \mathbb{R}$ satisfying local constancy along $\mathbb{R}$ is the same thing as an $\mathsf{SC}^{\mathrm{ch,top}}$-algebra.

The proof uses the product Weiss topology. A cover of $U \times I \subset \mathbb{C} \times \mathbb{R}$ is a collection of products $\{U_\alpha \times I_\beta\}$ refining the product open cover. The cosheaf of observables on $\mathbb{C} \times \mathbb{R}$ restricts to a constructible dg-cosheaf on $\mathrm{FM}_k(\mathbb{C}) \times \mathrm{Conf}_k(\mathbb{R})$. Weiss descent identifies the structure maps with the operations of $\mathsf{SC}^{\mathrm{ch,top}}$.

Theorem 2.1 *(Recognition; proved)*. There is an equivalence of ∞-categories between:
(i) logarithmic $\mathsf{SC}^{\mathrm{ch,top}}$-algebras, and
(ii) factorisation algebras on $\mathbb{C} \times \mathbb{R}$ that are holomorphic in $\mathbb{C}$ and locally constant in $\mathbb{R}$.

The product topology is not decorative. It is the entire content of the theorem: the two colours of the Swiss-cheese operad arise because the two factors of $\mathbb{C} \times \mathbb{R}$ carry different topologies (holomorphic versus locally constant).

The recognition theorem converts geometric data (a factorisation algebra on a 3-manifold) into algebraic data (an operad algebra). Without it, the Swiss-cheese structure would be a presentation, not a property. With it, the two-coloured operad is the geometry.

3 One algebra, three objects

From a single chiral algebra $\mathcal{A}$, the bar construction produces three objects. They are not the same.

(i) The **boundary algebra** $\mathcal{A}$ itself. This lives on the boundary $\{0\} \times \mathbb{R}$ of the half-space $\mathbb{C}_{\geq 0} \times \mathbb{R}$. It is a chiral algebra: a vertex algebra, an E_∞-chiral structure. Its data are the OPE coefficients c_n^{ab} and the conformal weights.

(ii) The **bar coalgebra** $\overline{B}(\mathcal{A})$. This is a factorisation coalgebra on $\mathrm{Ran}(X)$, carrying both the differential and the coproduct. It classifies *twisting morphisms*: universal couplings between $\mathcal{A}$ and its Koszul dual $\mathcal{A}^!$. On the

Koszul locus, $\mathrm{MC}(\mathrm{Hom}(\overline{B}(\mathcal{A}), \mathcal{A}^!)) \simeq \mathrm{Perf}(\mathcal{A}^!)$. This is the bar complex's native role: a representability theorem for deformations.

(iii) The **chiral derived center**

$$\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) \;:=\; H^*\big(C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A}),\, \delta\big).$$

This is the universal bulk algebra: it classifies observables of the 3d theory that act on the boundary. At the level of Hochschild cohomology: $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) \simeq \mathrm{HH}^\bullet_{\mathrm{ch}}(\mathcal{A})$, the chiral Hochschild cochains.

*

The identification "bar = bulk" is the most dangerous false equation in three-dimensional holomorphic-topological field theory.

The bar coalgebra classifies *couplings* between a boundary and its dual. The derived center classifies *operators* that the boundary does not know it has. These are different functors applied to different inputs with different outputs. In a sentence: bar-cobar is about twisting morphisms; the derived center is about the acting-on relation.

Object	**Classifies**	**Functor**
$\overline{B}(\mathcal{A})$	twisting morphisms $\mathcal{A} \rightsquigarrow \mathcal{A}^!$	bar (coalgebra)
$\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$	bulk operators acting on $\mathcal{A}$	chiral Hochschild (algebra)
$\Omega(\overline{B}(\mathcal{A}))$	nothing new: $\mathcal{A}$ itself	cobar (recovers input)

The cobar of the bar is $\mathcal{A}$ (Theorem B of Volume I). The Verdier dual of the bar is $\overline{B}(\mathcal{A}^!)$ (Theorem A). These are three different operations, producing three different objects.

4 The Swiss-cheese pentagon

The $\mathsf{SC}^{\mathrm{ch,top}}$ operad has two colours: *closed* (holomorphic, on $\mathrm{FM}_k(\mathbb{C})$) and *open* (topological, on $\mathsf{E}_1(m)$). The structure maps are:

Map	**Inputs**	**Output**	**Interpretation**
$\mu_{k,0}$	k closed	closed	Gerstenhaber bracket
$\mu_{0,m}$	m open	open	A_∞ products
$\mu_{k,m}$	k closed, m open	open	brace operations
$\mu^{\mathrm{oc}}_{k,0}$	k open	closed	**EMPTY**
$\mu^{\mathrm{mix}}_{k,m}$	mixed	mixed	SC composition

The fourth row is the content. No open inputs produce closed outputs. This is not a convention but a structural constraint: the open colour lives on the boundary $\mathbb{R}$, the closed colour in the bulk $\mathbb{C} \times \mathbb{R}$, and bulk interactions restrict to boundaries but not conversely. Bulk-to-boundary is a restriction functor; boundary-to-bulk would require extension, which the holomorphic direction forbids.

The bar complex makes this visible. The bar *differential* (closed colour) is the OPE residue extraction: a collision in the $\mathbb{C}$-plane. The bar *coproduct* (open colour) is the deconcatenation: a separation along $\mathbb{R}$. Applying

the coproduct and then the differential is a brace operation: splitting a bar element, then colliding within each piece. Applying the differential and then the coproduct is an SC composition: colliding first, then splitting. The Stasheff pentagon for the A_∞ structure and the Gerstenhaber bracket for the closed colour are the same identity, read on the two factors of $\mathrm{FM}_k(\mathbb{C}) \times \mathsf{E}_1(m)$.

The Swiss-cheese operad is Koszul.

THEOREM 4.1 *(Homotopy-Koszulity; proved).* The chiral-topological Swiss-cheese operad $\mathsf{SC}^{\mathrm{ch,top}}$ is homotopy-Koszul.

The proof: (1) the classical Swiss-cheese operad is Koszul (Livernet, Voronov, Ginzburg–Kapranov); (2) Kontsevich formality gives a quasi-isomorphism $\Phi\colon \mathsf{SC}^{\mathrm{ch,top}} \to \mathrm{SC}$; (3) bar-cobar preserves quasi-isomorphisms; two-out-of-three.

The consequences are unconditional. Bar-cobar on $\mathsf{SC}^{\mathrm{ch,top}}$-algebras is a Quillen equivalence, not merely an adjunction. Lines are $\mathcal{A}^!$-modules. The dg-shifted Yangian is defined without hypothesis.

5 Descent to the classical shadow

The holomorphic Swiss-cheese algebra, projected to cohomology, produces a Poisson vertex algebra (PVA). The descent is controlled by six axioms D1–D6, each proved by a geometric argument on $\mathrm{FM}_k(\mathbb{C})$.

D1. *Leibniz rule*: $\{a_\lambda bc\} = \{a_\lambda b\}c + b\{a_\lambda c\}$.
D2. *Sesquilinearity*: $\{\partial a_\lambda b\} = -\lambda\{a_\lambda b\}$.
D3. *Skew-symmetry*: $\{b_\lambda a\} = -\{a_{-\lambda-\partial} b\}$.
D4. *Jacobi identity*: $\{a_\lambda\{b_\mu c\}\} - \{b_\mu\{a_\lambda c\}\} = \{\{a_\lambda b\}_{\lambda+\mu} c\}$.
D5. *Commutativity*: $ab = ba$ on cohomology.
D6. *Unit*: $\{a_\lambda 1\} = 0$.

D2–D5 are proved by the exchange cylinder and the three-face Stokes identity on $\mathrm{FM}_3(\mathbb{C})$. The exchange cylinder is the bordism $\mathrm{FM}_2(\mathbb{C}) \times [0,1] \to \mathrm{FM}_3(\mathbb{C})$ that swaps two points while a third observes; the three-face identity

$$d\omega_3 \;=\; \omega_{12,3} - \omega_{13,2} + \omega_{1,23}$$

on $\partial\mathrm{FM}_3(\mathbb{C})$ gives skew-symmetry and Jacobi simultaneously. D6 uses the factorisation of $H_3(a)$ and the contractibility of the disc.

The bridge operates in both directions. F4 (operad $\Rightarrow$ axioms): the $\mathsf{SC}^{\mathrm{ch,top}}$ structure maps project to PVA operations. F5 (axioms $\Rightarrow$ operad): a PVA satisfying D1–D6 lifts to an $\mathsf{SC}^{\mathrm{ch,top}}$-algebra. The lift exists because $\mathsf{SC}^{\mathrm{ch,top}}$ is homotopy-Koszul: the cobar of the Koszul dual cooperad provides a cofibrant resolution, and the PVA data define a map from this resolution.

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The PVA on cohomology $H^\bullet(\mathcal{A}, Q)$ is (-1)-shifted: the λ-bracket has shifted parity relative to classical PVA conventions. This is not an anomaly. It is the degree shift in the desuspension s^{-1} of the bar construction.

6 The coproduct

In gauge theory (Chern–Simons with boundary algebra $\widehat{\mathfrak{g}}_k$), the gluon J^a has a primitive coproduct:

$$\Delta_z(J^a) \;=\; J^a(w+z) \otimes 1 \;+\; 1 \otimes J^a(w).$$

But the Sugawara composite $T_{\mathrm{aff}} = \frac{1}{2(k+h^\vee)} \sum_a :J^a J^a:$ has a nontrivial coproduct. The cross term

$$\Delta_z(T_{\mathrm{aff}}) \;=\; \tau_z(T_{\mathrm{aff}}) \otimes 1 \;+\; 1 \otimes T_{\mathrm{aff}} \;+\; \frac{1}{k+h^\vee}\,\Omega(w{+}z,\, w),$$

where $\Omega = \sum_a J^a \otimes J^a$ is the split Casimir, is a production vertex: a gluon bound state can dissociate into its constituents. The gluon IS the connection. It splits because the connection is a primitive field.

*

In gravity (Virasoro boundary), the stress tensor T is the graviton. The coproduct is primitive at every arity:

$$\Delta_z^W(x) \;=\; \tau_z(x) \otimes 1 \;+\; 1 \otimes x \tag{1}$$

for every element x of the $\mathcal{W}$-algebra, with no corrections whatsoever.

The mechanism is a ghost-number obstruction intrinsic to Drinfeld–Sokolov reduction. The transferred coproduct, computed by homological perturbation, sums over decorated trees. Every tree in the HPL expansion produces an output at ghost number ± 1. The projection $p \otimes p$ onto $\mathcal{W}$-algebra cohomology kills both. The obstruction is not perturbative: it holds at every order, every arity, every principal DS reduction in every simple Lie algebra.

In gauge theory, the gluon IS the connection. In gravity, the graviton IS the curvature. A connection splits (it is the primitive field of the theory). A curvature does not (it is a composite, and the ghost-number filtration forbids the splitting). This is the algebraic content of the statement that gravity does not produce particles; it only braids them.

	Chern–Simons	**Gravity**
Boundary algebra	$\widehat{\mathfrak{g}}_k$	Vir_c
Primitive field	J^a (connection)	T (curvature)
A_∞ tower	finite ($m_3 = 0$, class L)	infinite ($m_k \neq 0\ \forall k$, class M)
Coproduct Δ_z	nontrivial (Casimir cross-term)	primitive (ghost obstruction)
r-matrix	$\Omega_{\mathfrak{g}}/z$	$(c/2)/z^3 + 2T/z$
Shadow depth	3	∞

All nonlinearity of gravity resides in the A_∞ products $\{m_n^W\}_{n\geq 3}$ and in the collision residue $r^{\mathrm{Vir}}(z) \;=\; (c/2)/z^3 \;+\; 2T/z$. The coproduct carries no information. This is the central algebraic fact about three-dimensional quantum gravity.

7 Five worked examples

Chern–Simons

Gauge group G, level k. Boundary algebra: $\widehat{\mathfrak{g}}_k$. Class L, shadow depth 3. r-matrix: $r(z) \;=\; \Omega_{\mathfrak{g}}/z$ (Casimir, single pole). Coproduct: nontrivial (Casimir cross-term on composites). $\kappa = \dim(\mathfrak{g})(k+h^\vee)/(2h^\vee)$. Complementarity: $\kappa + \kappa^! = 0$ by Feigin–Frenkel.

The entire A_∞ structure is formal: $m_k = 0$ for $k \geq 3$ on cohomology. The bar complex is quadratic. The Koszul dual $\widehat{\mathfrak{g}}_{-k-2h^\vee}$ is another affine algebra at the reflected level. Lines: highest-weight modules for the dual. On the integrable locus ($k \in \mathbb{Z}_{\geq 0}$), the line category is $\mathrm{Rep}_q(\mathfrak{g})$ with $q \;=\; e^{\pi i/(k+h^\vee)}$, recoverable by the Kazhdan–Lusztig equivalence.

Gravity

Boundary algebra: Vir_c. Class M, shadow depth ∞. r-matrix: $r(z) = (c/2)/z^3 + 2T/z$ (two poles, quasi-trigonometric). Coproduct: primitive. $\kappa = c/2$; $\kappa^! = (26-c)/2$; $\kappa + \kappa^! = 13 \neq 0$.

The quartic OPE pole forces an infinite A_∞ tower. Every m_k is nonzero for generic c, computed by a wheel diagram on K_k with $k-1$ cubic vertices. The quartic contact invariant $Q^{\mathrm{ct}}_{\mathrm{Vir}} = 10/[c(5c+22)]$ has poles at $c = 0$ and $c = -22/5$: the Kac determinant zeros at weight 4. Self-duality at $c = 13$, not $c = 26$.

Twisted holography

Costello–Gaiotto. Boundary algebra: $V_k(\mathfrak{g})$ (affine Kac–Moody). Class L, shadow depth 3. Coproduct: nontrivial. The holographic dictionary is the Koszul triangle read through Volume I's five theorems: bar complex exists (Theorem A), boundary reconstructs bulk (Theorem B), Neumann/Dirichlet decomposition (Theorem C), genus tower by κ (Theorem D), BV origin of Hochschild complex (Theorem H).

M2-brane

From the $\mathcal{N} = 6$ ABJM theory. Boundary algebra: supersymmetric current algebra. Coproduct: nontrivial (fundamental fields split). The M2 example has no DS reduction: the $\mathcal{W}$-algebra structure is absent, and the coproduct obstruction does not apply. This is a gauge theory, not a gravity theory, and the coproduct confirms it.

M5-brane / 6d $(2,0)$

From the $\mathcal{N} = (2,0)$ theory on $\mathbb{C} \times \mathbb{R} \times S^3$. Boundary algebra: $\mathcal{W}_N$ at large N. Class M, shadow depth ∞. Coproduct: primitive (by the same ghost-number argument as Virasoro, since $\mathcal{W}_N$ arises by DS reduction).

	CS	Gravity	Twisted hol.	M2	M5
Class	L	M	L	L	M
$r_{\max}$	3	∞	3	3	∞
Δ_z	nontrivial	primitive	nontrivial	nontrivial	primitive
DS origin	no	yes	no	no	yes
$\kappa + \kappa^!$	0	13	0	0	ϱK_N

The dichotomy is clean: DS reduction produces primitive coproducts and infinite shadow depth. Its absence permits nontrivial coproducts and finite termination.

8 The annulus

Cut the circle S^1 by two arcs. Each arc is an interval $I \cong \mathbb{R}$. The factorisation homology of the open-sector category $\mathcal{C}_{\mathrm{op}}$ on S^1 is computed by excision:

$$\int_{S^1} \mathcal{C}_{\mathrm{op}} \;\simeq\; \mathcal{C}_{\mathrm{op}}(I_1) \otimes^{\mathbb{L}}_{\mathcal{C}_{\mathrm{op}}(I_1 \cap I_2)} \mathcal{C}_{\mathrm{op}}(I_2).$$

The two-arc cover produces the two-sided bar complex, which is the cyclic bar complex. In the homotopy category:

$$\int_{S^1} \mathcal{C}_{\mathrm{op}} \;\simeq\; \mathrm{HH}_*(\mathcal{C}_{\mathrm{op}}). \tag{2}$$

This is the annulus trace. The annulus $S^1 \times [0,1]$ is the simplest bordered surface with one boundary component. Its factorisation homology is the Hochschild homology of the open-sector category. The trace is the clutching operation: it restores the tangential basepoint τ_p to close the interval I_p back to a circle.

*

For affine Kac–Moody at integrable level k: $\mathrm{HH}_0(\mathcal{C}_{\mathrm{op}})$ is spanned by the characters $\chi_j(\tau) = \mathrm{tr}_{L_j}(q^{L_0 - c/24})$ for $j = 0, 1, \ldots, k$. The modular S-matrix acts on this space. Modular invariance of the partition function $Z(\tau) = \sum_j |\chi_j|^2$ is the statement that the trace is $SL(2,\mathbb{Z})$-equivariant.

The annulus trace is the **first modular shadow of the open sector**. It is the genus-1 datum, the simplest case of the trace-and-clutching construction that produces genus-g amplitudes from the open category. Modularity is not an axiom on the closed algebra. It is a property of the trace on the open sector.

9 The line category

On the chirally Koszul locus:

$$\mathcal{C}_{\mathrm{line}} \simeq \mathcal{A}^!\text{-}\mathbf{mod}. \tag{3}$$

Line operators are modules for the Koszul dual.

The proof is Koszul duality on the open colour. The bar complex $\overline{B}(\mathcal{A})$ is a dg coalgebra; its linear dual $\overline{B}(\mathcal{A})^\vee = \overline{B}(\mathcal{A}^!)$ (Verdier duality, Theorem A of Volume I) is a dg algebra. Maurer–Cartan deformations of the universal twisting morphism $\alpha\colon \overline{B}(\mathcal{A}) \to \mathcal{A}^!$ are in bijection with $\mathcal{A}^!$-module structures:

$$\mathrm{MC}\big(\mathrm{Hom}(\overline{B}(\mathcal{A}), \mathcal{A}^!)\big) \simeq \mathrm{Perf}(\mathcal{A}^!).$$

A line is not a module that happens to be placed along $\mathbb{R}$. A module is a line: the representation category of $\mathcal{A}^!$ IS the category of boundary conditions for the theory restricted to $\mathbb{R}$.

*

The meromorphic tensor structure on $\mathcal{C}_{\mathrm{line}}$ comes from the r-matrix. Two line operators at spectral parameters z_1 and z_2 tensor along the open direction with braiding

$$R(z_1 - z_2)\colon M_1 \otimes M_2 \xrightarrow{\sim} M_2 \otimes M_1,$$

where $R(z)$ is the quantum R-matrix built from the bulk-boundary composition (not from a quantum group, though they agree on the evaluation locus). For Chern–Simons: $R(z) = 1 + \hbar\,\Omega_{\mathfrak{g}}/z + O(\hbar^2)$. For gravity: $R(z) = 1 + \hbar\,((c/2)/z^3 + 2T/z) + O(\hbar^2)$.

The r-matrix is the genus-0, arity-2, ordered shadow of $\Theta_{\mathcal{A}}$:

$$r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}}).$$

Volume I computes the collision residue. Volume II reads the result as a spectral braiding on lines.

10 The benchmark

$V_k(\mathfrak{sl}_2)$ on $(\mathbb{P}^1, \{0, \infty\})$. This is the simplest non-trivial example: one curve with two punctures, one simple Lie algebra of rank 1, one level parameter k. Every object in the platonic datum can be computed.

Tangential log curve. $(X, D, \tau) = (\mathbb{P}^1, \{0, \infty\}, \{\tau_0, \tau_\infty\})$ with tangential basepoints in the positive-real direction. The boundary intervals $I_0, I_\infty \cong \mathbb{R}$ are the time axes at each puncture.

Open sector. $C_{\mathrm{op}} = V_k(\mathfrak{sl}_2)$-mod at each puncture. At integrable level k, this is $\mathrm{Rep}_q(\mathfrak{sl}_2)$ with $q = e^{i\pi/(k+2)}$, a semisimple category with $k+1$ simple objects $L_0, L_1, \ldots, L_k$.

Boundary algebra. $A_b = V_k(\mathfrak{sl}_2)$. The λ-bracket: $\{J^a{}_\lambda J^b\} = f^{ab}{}_c J^c + k\kappa^{ab}\lambda$.

Koszul dual. $V_k(\mathfrak{sl}_2)^! = V_{-k-4}(\mathfrak{sl}_2)$. The Feigin–Frenkel involution $k \mapsto -k - 2h^\vee = -k - 4$.

r-matrix. $r(z) = \Omega_{\mathfrak{sl}_2}/z = (J^a \otimes J_a)/z$, a single-pole solution of the classical Yang–Baxter equation.

Modular characteristic. $\kappa = 3(k+2)/4$. Complementarity: $\kappa + \kappa^! = 0$.

Derived center. $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{sl}_2)) \simeq \mathrm{HH}^\bullet_{\mathrm{ch}}(V_k(\mathfrak{sl}_2))$. At the critical level $k = -2$: functions on $\mathfrak{sl}_2$-opers.

Line category. $C_{\mathrm{line}} \simeq V_{-k-4}(\mathfrak{sl}_2)$-mod. At integrable levels: $\mathrm{Rep}_{q^{-1}}(\mathfrak{sl}_2)$.

Annulus trace. $\mathrm{HH}_0(C_{\mathrm{op}}) = \bigoplus_{j=0}^k \mathbb{C} \cdot \chi_j$. The modular S-matrix:

$$S_{jj'} = \sqrt{\frac{2}{k+2}} \sin\left(\frac{\pi(j+1)(j'+1)}{k+2}\right).$$

At $k = 1$: $S = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, the Hadamard matrix.

Factorisation homology. $\int_{\mathbb{C}^\times} C_{\mathrm{op}} \simeq \mathrm{CB}_{0,2}(V_k(\mathfrak{sl}_2))$: genus-0 conformal blocks with two insertions. For simple modules L_j and $L_{j'}$: $\dim \mathrm{CB}_{0,2}(L_j, L_{j'}) = \delta_{j,j'}$.

Every entry is proved. The benchmark is the frame atom of Volume II: simple enough to compute completely, rich enough to exhibit every feature of the theory.

11 Modularity last

The standard approach to modular invariance begins with the closed-string partition function and imposes modular invariance as an axiom. The Swiss-cheese framework reverses this.

Modularity is not an axiom on the closed algebra. It is a *derived property* of the trace on the open sector.

*

The modular Swiss-cheese operad extends $\mathsf{SC}^{\mathrm{ch,top}}$ by replacing trees with stable graphs carrying two edge colours: closed edges (genus direction, carrying $\hbar$-weight, recording non-separating degenerations) and open edges (boundary direction, carrying insertion data, recording separations along $\mathbb{R}$).

The relative Feynman transform $F_{\mathrm{rel}}\mathcal{O}$ sums over stable graphs with both edge colours. The differential contracts one edge; $d^2 = 0$ is the codimension-2 boundary cancellation on the moduli space of bordered Riemann surfaces.

At genus $g \geq 1$, the curvature $d_{\mathrm{fib}}^2 = \kappa \cdot \omega_g$ forces a passage from ordinary to curved Swiss-cheese algebras. The modular MC element

$$\Theta_{\mathcal{A}}^{\mathrm{oc}} \;:=\; \Theta_{\mathcal{A}} + \sum_j \mu^{\mathcal{M}_j} \tag{4}$$

packages both the bar-intrinsic MC element $\Theta_{\mathcal{A}}$ (from Volume I) and the A_∞-module structures $\mu^{\mathcal{M}_j}$ at punctures. Its MC equation decomposes into four components:

pure closed:	$d_0\,\Theta + \frac{1}{2}[\Theta,\Theta] = 0$
pure open:	$\partial\mu + \mu \circ \mu = 0$
mixed:	$d_0\,\mu + [\Theta,\mu] + \cdots = 0$
clutching:	(genus recursion)

The clutching component is the genus recursion from Volume I. Each genus is determined by all lower genera. No parameter is free.

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The annulus trace (the S^1 factorisation homology of C_{op}) produces the genus-1 modular datum. The clutching of g annuli produces the genus-g datum. Modularity emerges from the composition law of traces, not from an a priori constraint on the closed algebra.

The programme status:

Result	**Status**
Annulus trace $\simeq \mathrm{HH}_*(C_{\mathrm{op}})$	proved
Full modular cooperad	programme (in progress)
Modular Swiss-cheese $d^2 = 0$	proved (by $\partial^2 = 0$ on $\overline{\mathcal{M}}_{g,n}$)
Curvature $= \kappa \cdot \omega_g$	proved (Volume I)

12 The gravitational coproduct is primitive

THEOREM 12.1 *(Gravitational primitivity; proved).* For every principal Drinfeld–Sokolov reduction of every simple Lie algebra, the transferred dg-shifted Yangian coproduct on the $\mathcal{W}$-algebra is strictly primitive:

$$\Delta_z^W(x) \;=\; \tau_z(x) \otimes 1 \;+\; 1 \otimes x$$

for every x, at every arity, with no corrections.

Proof. Three lines.

The homological perturbation lemma expresses the transferred coproduct as a sum over decorated trees. Each tree takes $\mathcal{W}$-algebra inputs (ghost number 0), applies the BRST homotopy h (which raises or lowers ghost number by 1), and outputs a tensor product. Every tree with at least one internal vertex produces an output at ghost number ± 1. The projection $p \otimes p$ onto ghost-number-0 cohomology annihilates both signs. The only surviving contribution is the trivial tree (no internal vertices), which gives the primitive term $\tau_z(x) \otimes 1 + 1 \otimes x$. □

*

The proof works for every principal DS reduction because the ghost-number obstruction is intrinsic to the BRST structure, not to the specific Lie algebra. The ghost field $c \in \mathfrak{n}_+^*$ has ghost number $+1$; the antighost $b \in \mathfrak{n}_+$ has ghost number -1; the BRST homotopy h carries ghost number ± 1; and the projection p annihilates everything outside ghost number 0.

The physical content: gravity does not produce particles. The gravitational coproduct is primitive because the graviton is a composite (the stress tensor, built from the Sugawara construction) and not a fundamental field. The ghost-number filtration, which is the algebraic encoding of the BV grading, enforces this at every order.

For Chern–Simons, the fundamental field J^a has ghost number 0, so the BRST homotopy produces ghost-number-0 outputs. The Casimir cross-term survives projection. The gluon splits.

The asymmetry between gauge theory and gravity is an asymmetry between fundamental and composite fields, detected by the ghost number.

13 The four-stage architecture

The open/closed programme has four stages. Each depends on its predecessor.

Stage 1: local one-colour. The input is a single chiral algebra $\mathcal{A}$. Outputs:

(a) The bar coalgebra $\overline{B}(\mathcal{A})$ exists as a factorisation coalgebra (Volume I, Theorem A).
(b) On the Koszul locus, MC couplings $\simeq \mathrm{Perf}(\mathcal{A}^!)$ (Koszul duality on the open colour).
(c) The bulk is the chiral derived center: $\mathcal{Z}_{\mathrm{ch}}^{\mathrm{der}}(\mathcal{A}) \simeq \mathrm{HH}_{\mathrm{ch}}^\bullet(\mathcal{A})$ (boundary-linear exact theorem).
(d) Braces: $C_{\mathrm{ch}}^\bullet(\mathcal{A}, \mathcal{A})$ carries a brace dg-algebra structure from the $\mathsf{SC}^{\mathrm{ch,top}}$ action.

No global factorisation or modularity is used. Everything here is the universal property of the bar construction and the homotopy-Koszulity of $\mathsf{SC}^{\mathrm{ch,top}}$. Proved.

Stage 2: the open primitive. The primitive datum is the open/closed factorisation dg-category $\mathcal{C}$ on a pointed curve with tangential basepoints (X, D, τ).

(a) For any compact generator $b \in \mathcal{C}(J)$, the boundary algebra $A_b = \mathrm{RHom}_{\mathcal{C}(J)}(b, b)$ is an A_∞ chiral algebra.
(b) Different compact generators produce Morita-equivalent boundary algebras: the category $\mathcal{C}$ is the invariant, A_b is a chart.
(c) The bar complex $\overline{B}(A_b)$ encodes the twisting data of $\mathcal{C}$, and bar-cobar is the representability theorem for deformations of b inside $\mathcal{C}$.

Proved.

Stage 3: globalise. Replace $(X, D, \tau) = (\mathbb{C}, \{0\}, \tau_0)$ by a general tangential log curve. The real oriented blowup $\pi\colon \widetilde{X}_D \to X$ replaces each puncture p_i by a circle $S^1_{p_i}$; the tangential basepoint τ_{p_i} cuts this circle into an interval $I_{p_i} \cong \mathbb{R}$.

The mixed configuration spaces $\mathrm{Conf}^{\mathrm{oc}}_{I,\mathbf{m}}(X, D, \tau) = \mathrm{Conf}_I(X \setminus D) \times \prod_p \mathrm{Conf}^<_{m_p}(\mathbb{R})$ parametrise bulk insertions and boundary insertions together. The mixed Ran space is their colimit.

Local-global bridge: the local one-colour theorems of Stage 1 extend to global statements via descent along the mixed Ran space. Proved locally, the global extension requires compact generation and derived-center hypotheses.

Stage 4: modularise. The trace operation on $\mathcal{C}_{\mathrm{op}}$ (closing an interval to a circle) produces genus-g data from genus-0 data.

(a) The annulus trace $\mathrm{Tr}_{\mathcal{A}} \simeq \mathrm{HH}_*(C_{\mathrm{op}})$ is the first modular shadow. Proved.
(b) The full modular cooperad structure on the trace (the composition law for genus-g amplitudes via clutching of g handles) is a programme.

*

The architecture is sequential. Stage k depends on Stage $k-1$. One should never discuss Stage 4 before establishing Stage 1. The order is: objects first, then categories, then globalisation, then modularity.

14 The Koszul triangle

On the chirally Koszul locus, the corrected bulk-boundary-line triangle is:

$$A_{\mathrm{bulk}} \simeq \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(B_\partial) \simeq \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(C_{\mathrm{line}}) \simeq \mathrm{HH}^\bullet_{\mathrm{ch}}(\mathcal{A}^!), \qquad C_{\mathrm{line}} \simeq \mathcal{A}^!\text{-}\mathbf{mod}. \tag{5}$$

Three vertices:

- The **bulk** is $A_{\mathrm{bulk}} \simeq \mathcal{O}(T^*[-1]\mathcal{L})$: polyvector fields on the Lagrangian. Note: the restriction $\mathcal{O}(\mathcal{M})|_{\mathcal{L}} = \mathcal{O}(\mathcal{L})$ is the boundary, not the bulk. The bulk retains the full (-1)-shifted cotangent fibre.
- The **boundary** is $B_\partial = \mathcal{O}(\mathcal{L})$: functions on the Lagrangian.
- The **line category** is $C_{\mathrm{line}} = \mathfrak{S}_b$-mod: modules for the self-intersection algebra $\mathfrak{S}_b = \mathcal{L} \times_{\mathcal{M}} \mathcal{L}$.

The equivalence $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(B_\partial) \simeq A_{\mathrm{bulk}}$ is the Hochschild–Kostant–Rosenberg theorem: Hochschild cochains of $\mathcal{O}(\mathcal{L})$ are polyvector fields on $T^*[-1]\mathcal{L} \simeq \mathcal{M}|_{\mathcal{L}}$, by the Lagrangian condition $T_{\mathcal{L}}\mathcal{M}/T\mathcal{L} \simeq T^*\mathcal{L}[-1]$.

The equivalence $C_{\mathrm{line}} \simeq \mathcal{A}^!$-mod is the convolution-module model: the self-intersection algebra on $\mathfrak{S}_b$ is the Koszul dual, and its modules are lines.

Both equivalences require the Lagrangian condition. This is the geometric origin of the algebraic fact that the boundary is NOT the bulk and lines are NOT bulk modules.

*

The bar complex presents the Swiss-cheese algebra, as the Steinberg variety presents the Hecke algebra.

This is not analogy. Both are algebro-geometric presentations of operadic structures via configuration-space resolutions. The Steinberg variety $\mathfrak{S}_b = \mathcal{L} \times_{\mathcal{M}} \mathcal{L}$ is the self-intersection of the Lagrangian $\mathcal{L}$ in the moduli stack $\mathcal{M}$; its convolution algebra is the Hecke algebra. The bar complex $\overline{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with differential from collision residues and coproduct from deconcatenation is the Swiss-cheese algebra on $\mathrm{FM}_k(\mathbb{C}) \times \mathrm{Conf}_k(\mathbb{R})$; its convolution with the Koszul dual is the line category. The two become the same construction at the level of factorisation homology.

15 What we do not know

Three falsification tests

Test 1. The global bulk-boundary-line triangle (equation (5)) holds unconditionally in the boundary-linear exact sector. Whether it extends to all HT theories depends on two unverified hypotheses: compact generation

of the open-sector category, and the derived center being a quasi-isomorphism to the bulk. These hypotheses are verified for Chern–Simons and for inputs from the Costello–Davison–Gaiotto framework. They are not verified for gravity. A counterexample would mean that the bulk of a gravitational theory is not the derived center of its boundary: the simplest open/closed identification would fail.

Test 2. The gravitational primitivity theorem (§12) is proved for principal DS reductions. For non-principal DS reductions (hook-type in type A, arbitrary nilpotent f in general), the ghost-number argument applies verbatim, but the coproduct on the BRST cohomology has not been computed. Hook-type in type A is the proved corridor. Full non-principal primitivity is conjectural.

Test 3. The modular cooperad structure on the annulus trace (Stage 4 of the architecture) is a programme, not a theorem. The annulus trace itself is proved: $\mathrm{Tr}_{\mathcal{A}} \simeq \mathrm{HH}_*(\mathcal{C}_{\mathrm{op}})$. The full cooperad structure (clutching laws for genus $g \geq 2$, compatibility with the modular operad on $\overline{\mathcal{M}}_{g,n}$) requires the bordered FM compactification and its log-FM extension, which is the frontier of Mok's programme.

Programme status ledger

Component	What it requires	Status
$\mathsf{SC}^{\mathrm{ch,top}}$ homotopy-Koszulity	Kontsevich + Livernet	GREEN
Recognition theorem	Product Weiss descent	GREEN
PVA descent D1–D6	FM Stokes calculus	GREEN
Gravitational primitivity	Ghost-number obstruction	GREEN
Bulk = derived center (boundary-linear)	Stage 1 local theorem	GREEN
Bulk = derived center (global)	Compact generation	AMBER
Lines = $\mathcal{A}^!$-mod	Koszul locus	GREEN
Annulus trace	S^1 excision	GREEN
Full modular cooperad	Bordered FM + log extension	AMBER
BV/BRST = bar at $g \geq 1$	Genus-g QME	RED

Five false ideas to dismiss

1. "The bar complex is the bulk algebra." The bar complex classifies twisting morphisms. The bulk is the derived center. These are different objects produced by different functors.
2. "The boundary algebra determines the theory." The boundary algebra is a chart of the open-sector category, not its essence. Different compact generators produce Morita-equivalent charts. The category is the invariant.
3. "Modularity is a property of the closed algebra." It is a property of the trace on the open sector. The closed algebra inherits modularity from the clutching operation, not the reverse.
4. "Gravity and gauge theory differ only in the central charge." They differ in the coproduct. Gauge theory has nontrivial Δ_z; gravity has primitive Δ_z. The ghost-number obstruction is structural, not parametric.
5. "The Koszul dual lives in the bulk." It lives on the boundary: $\mathcal{A}^!$ is the boundary algebra for line operators, not a bulk observable.

16 The completed platonic datum

From a single chiral algebra $\mathcal{A}$, the completed platonic datum assembles eight components:

$$\Pi_X^{\mathrm{oc}}(\mathcal{A}) \;=\; \big(\overline{B}_X,\;\; \Omega_X,\;\; \Theta_{\mathcal{A}},\;\; \mathcal{Z}_{\mathrm{ch}}^{\mathrm{der}},\;\; \mathrm{Tr}_{\mathcal{A}},\;\; \mathrm{HH}_*,\;\; \mathfrak{R}_\bullet^{\mathrm{oc}},\;\; \nabla^{\mathrm{hol}}\big).$$

Component	Object	Role
$\overline{B}_X$	bar coalgebra	classifies twisting morphisms
Ω_X	cobar algebra	recovers $\mathcal{A}$ (Theorem B)
$\Theta_{\mathcal{A}}$	universal MC class	organises nonlinear tower
$\mathcal{Z}_{\mathrm{ch}}^{\mathrm{der}}$	chiral derived center	universal bulk
$\mathrm{Tr}_{\mathcal{A}}$	annulus trace	first modular shadow
HH_*	Hochschild homology	factorisation on S^1
$\mathfrak{R}_\bullet^{\mathrm{oc}}$	nonlinear resonance tower	quartic and higher obstructions
∇^{hol}	holographic connection	genus-g transport

The first six components are proved. The nonlinear resonance tower is proved at finite arity (Volume I, shadow Postnikov tower). The holographic connection is defined but its full modular extension is the frontier.

*

The completed platonic datum is the constructive endpoint of the theory. Given any chiral algebra, compute Π_X^{oc}. Theorems A–H of Volume I are structural properties of Π_X^{oc}. The five worked examples of §7 are instances of this computation.

The relationship between the six-fold platonic package $\Pi_X(L)$ of Volume I (built from a Lie conformal algebra L) and the eight-fold $\Pi_X^{\mathrm{oc}}(\mathcal{A})$ (built from a chiral algebra $\mathcal{A}$) is the factorisation envelope: the functor $L \mapsto U_X(L)$ that produces the enveloping vertex algebra from L. The six-fold package lives at the input; the eight-fold package lives at the output. The two additional components (derived center and annulus trace) are intrinsically 3-dimensional: they require the second direction.

17 From one quartic pole

The Virasoro λ-bracket

$$\{T_\lambda T\} \;=\; \partial T + 2T\lambda + \frac{c}{12}\,\lambda^3$$

has three terms: the simple pole (∂T, the derivative), the double pole ($2T\lambda$, the stress tensor), and the quartic pole ($\frac{c}{12}\lambda^3$, the central charge).

The quartic pole is the engine. It forces non-associativity of m_2. Non-associativity of m_2 forces m_3 by the Stasheff identity $\partial m_3 + m_2 \circ m_2 = 0$. Each m_k forces m_{k+1}. The A_∞ structure is genuinely infinite (class M, shadow depth ∞).

The collision residue of the λ-bracket is the gravitational r-matrix:

$$r(z) \;=\; (c/2)/z^3 + 2T/z.$$

Two poles: a triple pole from the central charge, a simple pole from the stress tensor. The pole orders are one less than the OPE (the d log kernel absorbs one power). The r-matrix has no even-order poles for a bosonic algebra.

*

From this single input:

(I) The A_∞ tower: $m_3, m_4, \ldots$, all nonzero for generic c. The quartic contact invariant $Q^{\mathrm{ct}} = 10/[c(5c + 22)]$.

(II) The Koszul dual: Vir_{26-c}. Self-duality at $c = 13$. The depth-zero resonance shadow.

(III) The genus tower: $\sum_{g\geq 1} F_g\, \hbar^{2g} = (c/2)(\hat{A}(i\hbar) - 1)$, with $\kappa_{\mathrm{eff}} = (c - 26)/2$.

(IV) The Koszul triangle: boundary Vir_c, bulk $\mathbb{C}[[c]]$, lines modelled by Vir_{26-c}-modules.

(V) The gravitational coproduct: primitive (ghost-number obstruction from DS reduction of $\widehat{\mathfrak{g}}_k$).

The transgression algebra of Volume I's working notes completes the picture. The Koszul dual $B = \mathrm{Vir}_{26-c}$ has curvature $\lambda = \kappa(B) \cdot \omega_1 = \frac{26-c}{2}\,\omega_1$. The transgression algebra

$$B_\lambda \;=\; B\langle\eta\rangle \,/\, (\eta^2 - \lambda), \qquad |\eta| = 1,$$

adjoins a Clifford generator. The secondary anomaly $u = \eta^2 = \lambda$ is linear in κ (a class, not a scalar). The gravity dichotomy:

	$c \neq 26$	$c = 26$
u	$\neq 0$	$= 0$
Clifford algebra	matrix (2^g-dim)	exterior (2^{2g}-dim)
Regime	perturbative	topological
Genus tower	resummable ($\hat{A}$-genus)	collapsed
Surface topology	Morita-trivial	genuine

The MC recursion determines the genus-g partition function from the genus-0 λ-bracket. No sum over topologies. No free parameters. The Maloney–Witten problem of enumeration (which topologies to include) is replaced by a problem of algebra (which Maurer–Cartan equation to solve).

Question 17.1. Does the graded character $\hat{Z}_1^{\mathrm{grav}}(c;\,\tau) := \chi_{\mathrm{gr}}(\mathfrak{G}_1^{\mathrm{grav}}(c);\,q)$ have non-negative Fourier coefficients for all $c > 1$?

18 The Steinberg principle

The bar complex presents the Swiss-cheese algebra, as the Steinberg variety presents the Hecke algebra.

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The Steinberg variety $\mathfrak{S} = \widetilde{\mathcal{N}} \times_{\mathcal{N}} \widetilde{\mathcal{N}}$ is the self-intersection of the Springer resolution. Its Borel–Moore homology carries the Hecke algebra by convolution. The Kazhdan–Lusztig basis arises from intersection cohomology of orbit closures. The representation theory of $\mathfrak{S}$ is the representation theory of the Hecke algebra: simple modules correspond to irreducible representations, standard modules to Verma modules, the duality functor to the Kazhdan–Lusztig involution.

The bar complex of a chiral algebra $\mathcal{A}$ lives on the product $\mathrm{FM}_k(\mathbb{C}) \times \mathrm{Conf}_k(\mathbb{R})$. The differential extracts collision residues on $\mathrm{FM}_k(\mathbb{C})$; the coproduct deconcatenates on $\mathrm{Conf}_k(\mathbb{R})$. The Koszul dual $\mathcal{A}^!$ is the linear dual of $\overline{B}(\mathcal{A})$ (Verdier duality). Line operators are $\mathcal{A}^!$-modules. The representation theory of $\overline{B}(\mathcal{A})$ is the representation theory of the boundary: MC couplings are line defects.

Steinberg	**Swiss-cheese**
$\widetilde{\mathcal{N}} \to \mathcal{N}$ (resolution)	$\mathrm{FM}_k(X) \to \mathrm{Conf}_k(X)$ (compactification)
$\mathfrak{S} = \widetilde{\mathcal{N}} \times_{\mathcal{N}} \widetilde{\mathcal{N}}$ (self-intersection)	$\overline{B}(\mathcal{A}) \otimes_{\overline{B}(\mathcal{A}\text{-bimod})} \overline{B}(\mathcal{A})$ (Hochschild)
$H_*^{\mathrm{BM}}(\mathfrak{S})$ (Hecke algebra)	$\mathcal{Z}_{\mathrm{ch}}^{\mathrm{der}}(\mathcal{A})$ (derived center)
KL basis (from IC)	PBW basis (from spectral sequence)
Hecke modules	$\mathcal{A}^!$-mod (line operators)
KL involution	Verdier/Koszul duality
Cell structure	Shadow depth classification (G/L/C/M)

The analogy becomes an identification at the level of factorisation homology. The factorisation homology of the bar complex on a closed surface is the convolution product on the self-intersection, which is the Hecke algebra. The two constructions produce the same algebra because both are computing the endomorphism algebra of a compact generator in a factorisation category on a configuration space. The Fulton–MacPherson compactification is the chiral analogue of the Springer resolution.

The cell structure of the Hecke algebra (Kazhdan–Lusztig cells, Lusztig's a-function, the asymptotic Hecke algebra) should have a shadow-tower analogue: the G/L/C/M classification of shadow depth is the analogue of the cell decomposition, with shadow depth playing the role of Lusztig's a-function. Whether this analogy extends to a theorem is open.

19 What remains

Everything in §1–§18 can be said in two sentences.

The bar complex of a chiral algebra on a curve carries a differential (holomorphic direction) and a coproduct (topological direction); together these form a Swiss-cheese algebra on $\mathrm{FM}_k(\mathbb{C}) \times \mathrm{Conf}_k(\mathbb{R})$, presenting the boundary theory of a 3d holomorphic-topological QFT whose bulk is the chiral derived center, whose lines are modules for the Koszul dual, and whose modularity arises from trace and clutching on the open sector. For gravity, the coproduct is primitive (ghost-number obstruction from DS reduction), so all nonlinearity resides in the A_∞ products and the collision residue $r(z)$; for gauge theory, the coproduct is nontrivial, and the full two-channel structure is active.

That is what the 1-form told us when we gave it a second direction. What follows is what it does when you push it further: into the bordered moduli space, along the modular cooperad, past the death of the derived category at genus 1, through the spectral Drinfeld strictification, into the gravitational dichotomy at $c = 26$. Each direction is a computation that is either finished or honestly open.

*

The following structures are proved:

1. The Swiss-cheese operad $\mathsf{SC}^{\mathrm{ch,top}}$ and its homotopy-Koszulity.
2. The recognition theorem (Weiss cosheaf descent).
3. PVA descent D1–D6 (exchange cylinder, Stokes).
4. The four-stage open/closed architecture (Stages 1–3 proved; Stage 4 programme).
5. The bulk-boundary-line triangle (boundary-linear sector proved; global conjectural).
6. Gravitational primitivity (principal DS; non-principal conjectural).
7. The benchmark: $V_k(\mathfrak{sl}_2)$ on $(\mathbb{P}^1, \{0, \infty\})$, every component computed.
8. The annulus trace $\simeq \mathrm{HH}_*(\mathcal{C}_{\mathrm{op}})$.
9. The completed platonic datum $\Pi_X^{\mathrm{oc}}(\mathcal{A})$.

The following structures are open:

1. The full modular cooperad on the trace.
2. The global bulk-boundary-line triangle outside the boundary-linear sector.
3. BV/BRST = bar at genus $g \geq 1$.
4. Non-principal gravitational primitivity beyond hook-type.
5. The spectral Drinfeld strictification for Kac–Moody algebras with root multiplicities > 1.

*

A 1-form with a simple pole.

A relation among three.

A second direction: the coproduct.

The rest follows.

20 The composite graviton

The Virasoro stress tensor

$$T \;=\; \frac{1}{2(k+h^\vee)} \sum_a :J^a J_a:$$

is built from affine currents. It is not fundamental. The graviton is composite.

The ghost sector of the BRST complex remembers this. The Sugawara construction passes through the normal ordering, which carries ghost number 0 on each input but produces an output whose BRST homotopy h necessarily shifts ghost number by ± 1. Every tree in the HPL expansion of the transferred coproduct has at least one internal vertex; every internal vertex passes through h. The projection $p \otimes p$ onto ghost-number-0 cohomology annihilates the result. The coproduct vanishes:

$$\Delta^{\mathrm{grav}}(x) \;=\; \tau_z(x) \otimes 1 \;+\; 1 \otimes x.$$

In gauge theory, the gluon IS the connection. J^a is fundamental: ghost number 0, no passage through h, no obstruction. The cross-term survives:

$$\Delta_z(T_{\mathrm{aff}}) \;=\; \tau_z(T_{\mathrm{aff}}) \otimes 1 \;+\; 1 \otimes T_{\mathrm{aff}} \;+\; \frac{1}{k+h^\vee}\,\Omega(w{+}z,\, w).$$

The Casimir $\Omega = \sum_a J^a \otimes J_a$ is a production vertex. A gluon bound state dissociates into its constituents.

Regime	Gauge	Gravity
Primitive field	J^a (connection)	T (curvature)
Origin	fundamental	Sugawara composite
Coproduct	nontrivial	primitive
Physical reading	particles split	only braids

The r-matrix $r^{\mathrm{Vir}}(z) = (c/2)/z^3 + 2T/z$ carries all gravitational scattering. The cubic pole: the bar kernel $d\log(z-w)$ absorbs one power from the quartic OPE. The logarithmic derivative is always one degree milder than the function itself. This absorption is the structural content of calling the bar construction a categorical logarithm.

21 The holographic triangle

Three vertices.

Boundary: $\mathcal{A}$ (the chiral algebra).
Bulk: $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = H^*(C^\bullet_{\mathrm{ch}}(\mathcal{A},\mathcal{A}),\ \delta)$ (the chiral derived center).
Lines: $\mathcal{A}^!$-modules (on the chirally Koszul locus).

*

The bar complex $\overline{B}(\mathcal{A})$ classifies couplings between boundary and lines. It is the universal twisting coalgebra: every map $\alpha\colon \overline{B}(\mathcal{A}) \to \mathcal{A}^!$ satisfying the twisting morphism equation is a line operator. On the Koszul locus:

$$\mathrm{MC}\big(\mathrm{Hom}(\overline{B}(\mathcal{A}),\ \mathcal{A}^!)\big) \ \simeq\ \mathrm{Perf}(\mathcal{A}^!).$$

This is the bar complex's role: a representability theorem. It classifies morphisms, not operators.

The derived center $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ classifies bulk observables. It acts on the boundary via the brace dg-algebra structure on $C^\bullet_{\mathrm{ch}}(\mathcal{A},\mathcal{A})$, encoding how bulk fields modify boundary correlators. The restriction $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) \to \mathcal{A}$ is well-defined. The extension $\mathcal{A} \to \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ is not: bulk restricts to boundary, boundary does not determine bulk.

Warning 21.1. $\overline{B}(\mathcal{A}) \neq \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$. The bar complex is a Fourier kernel (Volume I). The derived center is a space of endomorphisms. One classifies morphisms; the other classifies operators. Their conflation is the most frequent source of false statements in three-dimensional holomorphic-topological field theory.

22 Where modularity lives

Modularity is not an axiom on the closed algebra. It is trace plus clutching on the open sector.

*

Cut S^1 into two arcs. Each arc is an interval $I \cong \mathbb{R}$. Excision on the factorisation homology of C_{op}:

$$\int_{S^1} C_{\mathrm{op}} \;\simeq\; C_{\mathrm{op}}(I_1) \;\otimes^{\mathbb{L}}_{C_{\mathrm{op}}(I_1 \cap I_2)}\; C_{\mathrm{op}}(I_2).$$

The two-arc cover produces the two-sided bar complex, which is the cyclic bar complex. Hochschild homology $\mathrm{HH}_*(C_{\mathrm{op}})$ IS the annulus partition function.

The first modular shadow: $\kappa(\mathcal{A})$ appears as the leading coefficient of the annulus trace. The genus-1 curvature $d^2_{\mathrm{fib}} = \kappa \cdot \omega_1$ is the obstruction to extending the genus-0 Swiss-cheese structure to the torus. The obstruction lives in the Hochschild homology of the open sector. It does not live in any axiom on the closed algebra.

Clutching: the genus-g surface Σ_g is built from $3g-3$ pairs of pants, sewn along $3g-3$ circles. Each sewing circle contributes an HH_* factor. The modular partition function is the trace over all sewings:

$$Z_g \;=\; \mathrm{Tr}_{\mathrm{HH}_*^{\otimes(3g-3)}}\big(\text{clutching maps}\big).$$

This is modularity from geometry, not from axioms.

23 The Koszul dual lives on the boundary

$D_{\mathrm{Ran}}(\overline{B}(\mathcal{A})) \simeq \overline{B}(\mathcal{A}^!)$: the Verdier dual of the bar coalgebra is the bar of the Koszul dual algebra $\mathcal{A}^!$ (Theorem A intertwining). The Koszul dual $\mathcal{A}^!$ lives on the $\mathbb{R}$-direction (the boundary of the half-space $\mathbb{C}_{\geq 0} \times \mathbb{R}$), not in the $\mathbb{C} \times \mathbb{R}$ bulk.

*

For Chern–Simons with gauge group G at level k: $\mathcal{A} = V_k(\mathfrak{g}), \mathcal{A}^! = V_{-k-2h^\vee}(\mathfrak{g})$. The Feigin–Frenkel involution $k \mapsto -k-2h^\vee$ is Verdier duality on $\mathrm{Ran}(X)$. At the critical level $k = -h^\vee$: $V_{-h^\vee}(\mathfrak{g})$ is commutative, bar cohomology is functions on opers, and $\mathcal{A}^!$ at the dual critical level is the Feigin–Frenkel center.

For gravity: $\mathcal{A} = \mathrm{Vir}_c, \mathcal{A}^! = \mathrm{Vir}_{26-c}$.

At $c = 26$: $\mathcal{A}^! = \mathrm{Vir}_0$, uncurved ($\kappa = 0$).

At $c = 0$: $\mathcal{A} = \mathrm{Vir}_0$, uncurved.

At $c = 13$: $\mathcal{A} = \mathcal{A}^!$. Self-duality.

The complementarity sum:

$$\kappa(\mathrm{Vir}_c) + \kappa(\mathrm{Vir}_{26-c}) \;=\; \frac{c}{2} + \frac{26-c}{2} \;=\; 13 \;\neq\; 0.$$

For Kac–Moody algebras: $\kappa + \kappa^! = 0$ by Feigin–Frenkel (the involution $k \mapsto -k-2h^\vee$ is anti-symmetric around 0). For Virasoro: the involution $c \mapsto 26-c$ is anti-symmetric around 13, not around 0. The midpoint matters. It is 13, not 0.

24 The BTZ black hole from the r-matrix

The effective curvature of the matter-ghost system:

$$\kappa_{\text{eff}} \;=\; \kappa(\mathrm{Vir}_c) + \kappa_{\text{ghost}} \;=\; \frac{c}{2} - 13 \;=\; \frac{c-26}{2}.$$

At $c = 26$: $\kappa_{\text{eff}} = 0$. No curvature. The theory is topological.

At $c \neq 26$: the genus tower $F_g = \kappa_{\text{eff}} \cdot \lambda_g^{\text{FP}}$. The $\hat{A}$-genus generates all F_g from κ_{eff}:

$$\sum_{g\geq 1} F_g\, \hbar^{2g} \;=\; \kappa_{\text{eff}}\big(\hat{A}(i\hbar) - 1\big).$$

The Bernoulli numbers encode the higher-genus corrections. Each F_g is positive (because $\hat{A}(ix) = (x/2)/\sin(x/2)$ has all positive Taylor coefficients).

*

The collision residue $r^{\mathrm{Vir}}(z) = (c/2)/z^3 + 2T/z$ controls BTZ thermodynamics. The leading coefficient $c/2 = \kappa$ is the scalar that generates the Bekenstein–Hawking entropy at leading order in $1/G$. The subleading $2T/z$ is state-dependent braiding: it distinguishes different BTZ states within the same theory.

The primitive coproduct is the algebraic statement that black holes do not split. The only channel is braiding. The production vertex $\Omega(w+z, w)$ of gauge theory (the Casimir cross-term) is absent. There is no splitting channel.

	$c \neq 26$	$c = 26$
κ_{eff}	$(c-26)/2 \neq 0$	0
F_1	$(c-26)/48$	0
Genus tower	$\kappa_{\text{eff}} \cdot (\hat{A} - 1)$	collapsed
Regime	perturbative	topological

25 Five theories, one machine

	CS	Gravity	Twisted hol.	M2	M5
$\mathcal{A}$	$V_k(\mathfrak{g})$	Vir_c	$V_k(\mathfrak{g})$	DDCA_K	$\mathcal{W}_\infty(K)$
$\mathcal{A}^!$	$V_{-k-2h^\vee}(\mathfrak{g})$	Vir_{26-c}	$V_{-k-2h^\vee}(\mathfrak{g})$	CE	?
κ	$\dim(\mathfrak{g})\frac{k+h^\vee}{2h^\vee}$	$c/2$	$\dim(\mathfrak{g})\frac{k+h^\vee}{2h^\vee}$	$K(K{+}1)/2$	$\varrho_\infty c$
Class	L	M	L	L	M
Δ_z	nontrivial	primitive	nontrivial	nontrivial	primitive
$\kappa + \kappa^!$	0	13	0	0	ϱK_N

The machine does not know which theory it is computing. It receives: a chiral algebra, its OPE, and the bar construction. The five theories are five inputs to the same functor.

Chern–Simons and twisted holography are class L: the r-matrix carries all structure, the coproduct is nontrivial, the shadow tower terminates at arity 3. Gravity and M5 are class M: infinite shadow depth, primitive coproduct, $Q^{\text{ct}} \neq 0$.

*

The M2–M5 duality: the M2 algebra (DDCA, class L) and the M5 algebra ($\mathcal{W}_\infty$, class M) are related by the Miura transformation. DS reduction converts class L to class M. The reduction IS the depth increase. The ghost-number filtration that kills the coproduct of M5 is the same filtration that was absent for M2.

26 The four-stage architecture, revisited

Stage 1 (local one-colour): define the Swiss-cheese algebra. Prove the universal acting-bulk theorem $U(\mathcal{A}) = (C^*_{\mathrm{ch}}(\mathcal{A}, \mathcal{A}),\ \mathcal{A})$. Everything succeeds or fails here.

Stage 2 (open primitive): the primitive object is $\mathcal{C}_{\mathrm{op}}$, not A_b. The boundary algebra $A_b = \mathrm{End}(b)$ is a chart of a compact generator b in $\mathcal{C}_{\mathrm{op}}$. Different generators produce Morita-equivalent charts. The category is the invariant.

Stage 3 (globalise): move from $\mathbb{C} \times \mathbb{R}$ to tangential log curves (X, D, τ). The real oriented blowup $\pi\colon \widetilde{X}_D \to X$ replaces each puncture p_i by a circle $S^1_{p_i}$; the tangential basepoint τ_{p_i} cuts this circle into an interval $I_{p_i} \cong \mathbb{R}$. Open insertions live on the interval. Bulk insertions live in the interior. The global theory is a factorisation algebra on a bordered Riemann surface.

Stage 4 (modularise): introduce Θ^{oc}, traces, annulus/Hochschild chains, clutching, quartic resonance classes. Only after stages 1–3 are solid.

*

Each stage depends on its predecessor. Discussion of stage k before establishment of stage $k-1$ is not premature; it is wrong.

27 What the bar complex knows about quantum gravity

The bar complex $\overline{B}(\mathrm{Vir}_c)$ is a factorisation coalgebra on $\mathrm{Ran}(X)$. Its differential $d_{\overline{B}}$ extracts collision residues of the Virasoro OPE. Its coproduct Δ deconcatenates ordered tuples along $\mathbb{R}$.

At genus 0: $d_{\overline{B}}^2 = 0$ by Arnold. The bar complex is an honest chain complex. Its cohomology has dimensions

$$1,\ 2,\ 5,\ 12,\ 30,\ 76,\ 196,\ 512,\ 1353,\ \ldots$$

The generating function satisfies $x^3P^2 - (1-2x)(1-x^2)P + (1-x)(1-x^2) = 0$. Discriminant: $(1-x)^2(1-3x)(1+x)$. Growth rate 3.

*

At genus 1: $d_{\mathrm{fib}}^2 = (c/2) \cdot \omega_1 \neq 0$. The bar complex is curved. The curvature $c/2$ is the modular characteristic.

At genus g:

$$F_g = \frac{c}{2} \cdot \lambda_g^{\mathrm{FP}} = \frac{c}{2} \cdot \frac{(2^{2g-1}-1)\,|B_{2g}|}{(2g)! \cdot 2^{2g-1}}.$$

The Bernoulli numbers. The $\hat{A}$-genus. From index theory. Nothing in the Virasoro OPE predicts this. It lives in the geometry of $\overline{\mathcal{M}}_g$.

*

The shadow tower

$$\begin{aligned} S_2 &= c/2, \\ S_3 &= 2, \\ S_4 &= 10/\big[c(5c+22)\big], \\ S_5 &= -48/\big[c^2(5c+22)\big], \end{aligned}$$

and all higher S_r, are the Taylor coefficients of $\sqrt{Q}$ where

$$Q_{\mathrm{Vir}}(t) \;=\; c^2 + 12ct + \frac{180c+872}{5c+22}\,t^2.$$

One quadratic. One square root. An algebraic function of degree 2.

Three numbers determine the entire infinite tower of higher-genus corrections to quantum gravity: $\kappa = c/2$, $\alpha = 2$ (the cubic shadow), and $S_4 = 10/[c(5c+22)]$ (the quartic contact invariant). The shadow metric Q is constructed from these three. Its discriminant $\Delta = 8\kappa S_4$ controls the convergence radius of the shadow tower. When $\Delta \neq 0$ (all $c \neq 0$): the tower is infinite, class M. The bar complex knows the exact answer at every genus, as the coefficients of an algebraic function.

28 The arithmetic structure of quantum gravity

The gravitational shadow tower is the arithmetic of the quadratic extension $K = \mathbb{Q}(c)(t)(\sqrt{Q_{\mathrm{Vir}}})$ over the function field $F = \mathbb{Q}(c)(t)$. The shadow metric

$$Q_{\mathrm{Vir}}(t) \;=\; c^2 + 12ct + \frac{180c+872}{5c+22}\,t^2$$

is a quadratic polynomial whose square root generates the infinite tower of higher-genus corrections to three-dimensional quantum gravity.

*

The discriminant. $\mathrm{disc}(Q_{\mathrm{Vir}}) = -320\,c^2/(5c{+}22)$. Negative for all $c > 0$: the roots of Q_{Vir} are complex conjugate, the extension K/F is genuine, and the shadow tower is infinite (class M).

The discriminant controls three quantities at once:

(i) The depth of the gravitational perturbation series (infinite for all $c \neq 0$, because $\Delta = 40/(5c{+}22) \neq 0$).

(ii) The convergence radius of the shadow tower: the growth rate $\rho = \sqrt{36 + 80/(5c{+}22)}\,/\,c$ is the reciprocal of the distance from the origin to the nearest branch point of $\sqrt{Q_{\mathrm{Vir}}}$.

(iii) The complementarity with the dual theory at $c' = 26{-}c$: $\Delta(c){+}\Delta(26{-}c) = 6960/[(5c{+}22)(152{-}5c)]$, with universal numerator.

*

The gravitational number field. The Bekenstein–Hawking entropy $S_{\mathrm{BH}} = 2\pi\sqrt{ch/6}$ derives from $\kappa = c/2$, the leading coefficient of Q_{Vir}. This coefficient is the constant term (up to a square): $Q_{\mathrm{Vir}}(0) = c^2 = 4\kappa^2$. The leading OPE datum seeds the quadratic extension.

The growth rate ρ measures how fast the higher-genus corrections to the gravitational partition function grow. When $\rho < 1$, the shadow tower converges: the shadow partition function $Z^{\mathrm{sh}} = \sum Z_g^{\mathrm{sh}} \hbar^{2g}$ exists as a convergent series. This convergence is the "splitting" of the gravitational prime at central charge c: the quadratic extension has a rational point in a neighbourhood of the origin.

Setting $\rho = 1$ yields the critical cubic $5c^3 + 22c^2 - 180c - 872 = 0$ with positive root $c^* \approx 6.125$. For $c > c^*$: the shadow tower converges ($\rho < 1$, the gravitational perturbation series sums). For $0 < c < c^*$: the shadow tower diverges ($\rho > 1$, the perturbation series requires resummation). The critical cubic is the Minkowski bound of the gravitational number field: it separates the convergent regime from the divergent regime.

*

The three critical values.

c	**What happens**	**Arithmetic meaning**
0	$\kappa = 0$, Q degenerates	ramified prime (shadow tower is undefined)
13	$\mathcal{A} = \mathcal{A}^!$, self-dual	Galois involution is trivial on coefficients; $K = K'$
26	$\kappa_{\mathrm{eff}} = 0$, topological	dual extension K' degenerates ($\kappa' = 0$)

At $c = 0$: the constant term of Q_{Vir} vanishes. The quadratic polynomial has a double root at $t = 0$. The shadow connection ∇^{sh} has an irregular singularity. The extension ramifies.

At $c = 13$: the theory is self-dual ($\mathrm{Vir}_{13}^! = \mathrm{Vir}_{13}$). The Koszul involution $c \mapsto 26 - c$ fixes $c = 13$. The two quadratic extensions K and K' coincide. The complementarity sum $\kappa + \kappa' = 13$ does not vanish: self-duality is a symmetry enhancement, not anomaly cancellation. The growth rate $\rho(13) \approx 0.467$: the shadow tower converges at the self-dual point.

At $c = 26$: the effective curvature $\kappa_{\mathrm{eff}} = (c-26)/2 = 0$ vanishes. The dual algebra Vir_0 has $\kappa' = 0$: its shadow extension ramifies. The gravitational theory becomes topological. The Clifford algebra degenerates from a matrix algebra (2^g-dimensional) to an exterior algebra (2^{2g}-dimensional). The BTZ partition function collapses: $Z_{\mathrm{grav}} = 1$.

These three values are not phases of one theory. They are three qualitatively distinct points in the arithmetic landscape of the quadratic extension: a ramified prime ($c = 0$), a self-dual fixed point ($c = 13$), and a degeneration of the dual ($c = 26$). They are separated by 13 units of central charge.

*

The ramified primes. The denominator theorem (Volume I) gives $\mathrm{denom}(S_r) = (\mathrm{const}) \cdot c^{r-3}(5c+22)^{\lfloor (r-2)/2 \rfloor}$. Two primes ramify:

(i) $c = 0$: the leading OPE coefficient vanishes. No Bekenstein–Hawking entropy. No bar curvature. The shadow metric $Q(0,t) = 872\,t^2/(5 \cdot 22) = 872\,t^2/110$ degenerates.

(ii) $c = -22/5$: the weight-4 Gram determinant $\det G_4 = c^2(5c+22)/2$ vanishes. A null vector at weight 4 collapses the quasi-primary Λ onto $\partial^2 T$. The quartic contact invariant $Q^{\mathrm{ct}} = 10/[c(5c+22)]$ has a pole.

Every other value of c is unramified: the shadow coefficients are finite, the quadratic extension is well-defined, and the tower runs.

No higher Kac determinant appears. The gravitational number field has exactly two ramified primes, both visible from the weight-4 representation theory. The infinite tower of higher-genus gravitational corrections is controlled by a single 2×2 Gram matrix.

*

The unit group and the BTZ partition function. The growth rate $\rho = \sqrt{36 + 80/(5c+22)}\,/\,c$ is the fundamental unit of the gravitational number field. It controls the BTZ partition function through the radius of convergence of the shadow tower:

- $\rho < 1$ ($c > c^*$): the shadow tower converges. The gravitational path integral exists non-perturbatively as a convergent shadow series. The shadow partition function Z^{sh} is the value of a convergent double series (genus and arity).
- $\rho > 1$ ($0 < c < c^*$): the shadow tower diverges. The shadow coefficients grow exponentially. Resummation (Borel transform of $\sqrt{Q_{\mathrm{Vir}}}$) is required. The Stokes constants of the shadow tower are the non-perturbative content.
- $\rho = 1$ ($c = c^*$): the critical cubic. The branch point of $\sqrt{Q_{\mathrm{Vir}}}$ lies exactly on the unit circle. The shadow tower is marginally divergent.

c	ρ	**Regime**
1	2.99	divergent (shadow)
$c^* \approx 6.125$	1.00	critical
13	0.467	convergent
26	0.234	convergent (dual topological)
100	0.061	strongly convergent

The string partition function Z_{string} (genus factorial growth $(2g)!$) always diverges. The shadow partition function Z^{sh} (Bernoulli decay $1/(2\pi)^{2g}$ times shadow growth $\rho^r r^{-5/2}$) converges for $c > c^*$. The shadow tower is better-behaved than the string because it sums over arities within a fixed genus, while the string sums over genera with unrestricted growth.

*

The Galois involution. The monodromy -1 of the shadow connection $\nabla^{\mathrm{sh}} = d - Q'/2Q\,dt$ around each branch point of $\sqrt{Q_{\mathrm{Vir}}}$ is the nontrivial element of $\mathrm{Gal}(K/F)$. In the gravitational context: the Koszul sign is the Galois involution of the number field. The sign alternation in the shadow tower S_r (positive for even r, approximately negative for odd r, with beat period $\pi/(\pi - \theta)$ where $\theta = \arg(t_+)$) is the manifestation of the Galois action on the Taylor coefficients of $\sqrt{Q_{\mathrm{Vir}}}$.

This is the deepest structural content of the bar construction's desuspension: the sign $(-1)^n$ at bar degree n is the Galois automorphism of a quadratic extension, evaluated at the n-th coefficient.

*

The gravitational shadow tower is not like a quadratic extension. It is one, over the function field $\mathbb{Q}(c)(t)$ instead of $\mathbb{Q}$. The Bekenstein–Hawking entropy is the leading coefficient. The convergence of the perturbation series is the splitting behaviour. The Koszul sign is the Galois involution. The critical cubic is the Minkowski bound. The two ramified primes are the Kac determinant zeros. The complementarity identity is the duality constraint on discriminants. The gravitational partition function is the norm form of the extension, evaluated at a specific point.

A 1*-form with a simple pole.*

A relation among three.

A second direction: the coproduct.

A quadratic extension: the shadow.

The rest follows.

29 Gravity at rational central charge

The generic gravitational number field $K = \mathbb{Q}(c)(t)(\sqrt{Q_{\mathrm{Vir}}})$ of the preceding section lives over the function field $\mathbb{Q}(c)$. When the central charge specialises to a rational number c_0, the field specialises to a concrete quadratic extension of $\mathbb{Q}$.

For a minimal model $c_{p,q} = 1 - 6(p-q)^2/(pq)$, the critical discriminant $\Delta(c_{p,q}) = 40/(5c_{p,q} + 22)$ is a specific rational number, and $K(c_{p,q}) := \mathbb{Q}(\sqrt{\Delta(c_{p,q})})$ is a specific real quadratic number field. Gravity at each minimal model selects a number field.

*

The physical picture. The Bekenstein–Hawking entropy $\kappa = c/2$ at a rational central charge $c_{p,q}$ is a rational number. The higher-genus corrections $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ are rational multiples of κ, hence rational. The ENTIRE perturbative gravitational partition function $\sum F_g \hbar^{2g}$ has rational coefficients.

But the shadow tower is different. The Taylor coefficients S_r of $\sqrt{Q_L}$ are rational (they satisfy a rational recursion), yet the generating function $H(t) = t^2\sqrt{Q_L(t)}$ is genuinely irrational: it involves $\sqrt{\Delta}$, which lies outside $\mathbb{Q}$ for all $c > 0$. The irrationality is the arithmetic content of the shadow tower. The bar complex of the Virasoro algebra, specialised to a minimal model, produces an algebraic number that no finite truncation of the perturbative expansion can detect.

This is not a deficiency of perturbation theory. The function $H(t)$ is algebraic of degree 2; its Taylor coefficients are all individually rational. The irrationality is non-perturbative in the arity variable t, not in the genus variable $\hbar$. It is visible in the branch structure of $\sqrt{Q_L}$, not in any single coefficient.

*

Computation. Write $5c_0 + 22 = a/b$ in lowest terms. Then $\Delta(c_0) = 40b/a$, and the specialised field is $K(c_0) = \mathbb{Q}(\sqrt{\mathrm{sqfree}(40ab)})$.

$c = 1/2$ *(Ising model).* $5c + 22 = 49/2$. $\Delta = 80/49$. $\sqrt{80/49} = 4\sqrt{5}/7$. The Ising gravity lives over $\mathbb{Q}(\sqrt{5})$.

The golden ratio $(1 + \sqrt{5})/2$ is the fundamental unit of this field, with norm -1. The class number is $h = 1$: the ring of integers $\mathbb{Z}[(1 + \sqrt{5})/2]$ is a principal ideal domain. The Ising model produces the simplest possible arithmetic. One is tempted to say that the Ising model is "arithmetically trivial," as it is physically the simplest unitary minimal model.

The shadow growth rate is $\rho(1/2) \approx 12.53$: the shadow tower diverges violently ($c = 1/2$ is far below the critical cubic $c^* \approx 6.125$). The Ising gravity requires resummation.

$c = 7/10$ *(tricritical Ising).* $5c + 22 = 51/2$. $\Delta = 80/51$. $K(7/10) = \mathbb{Q}(\sqrt{255})$ where $255 = 3 \cdot 5 \cdot 17$. Discriminant 1020. Class number $h = 4$: a genuine failure of unique factorisation.

$c = 4/5$ *(three-state Potts).* $K(4/5) = \mathbb{Q}(\sqrt{65})$. Discriminant 65. $h = 2$. The Potts-model gravity has a non-trivial ideal class group $\mathrm{Cl}(K) \cong \mathbb{Z}/2$.

$c = 13$ *(self-dual).* $K(13) = \mathbb{Q}(\sqrt{870})$ where $870 = 2 \cdot 3 \cdot 5 \cdot 29$. Discriminant 3480. $h = 8$.

The self-dual gravity has the largest class number of any integer central charge in $[0, 26]$. It sees the prime 29, inherited from the complementarity constant $6960 = 2^4 \cdot 3 \cdot 5 \cdot 29$. No other central charge in the standard range detects this prime.

$c = 26$ *(critical string).* $K(26) = \mathbb{Q}(\sqrt{95})$. Discriminant 380. $h = 2$. $\rho(26) \approx 0.232$: convergent. This is the gravitational number field of the bosonic string at the critical dimension.

c	$\Delta(c)$	$d(c)$	disc	h	ε	$\rho(c)$
$1/2$	$80/49$	5	5	1	$(1+\sqrt{5})/2$	12.53
$7/10$	$80/51$	255	1020	4	$16+\sqrt{255}$	8.94
$4/5$	$20/13$	65	65	2	$8+\sqrt{65}$	7.81
1	$40/27$	30	120	2	$11+2\sqrt{30}$	6.24
13	$40/87$	870	3480	8	$59+2\sqrt{870}$	0.47
25	$40/147$	30	120	2	$11+2\sqrt{30}$	0.24
26	$5/19$	95	380	2	$39+4\sqrt{95}$	0.23

*

Koszul duality and the field pair. The Koszul pair $(c, 26 - c)$ produces two fields $K(c)$ and $K(26 - c)$. These coincide iff $(5c + 22)/(152 - 5c) \in \mathbb{Q}^{\times 2}$.

At $c = 1$ and $c = 25$: the ratio is $27/147 = (3/7)^2$. Both give $\mathbb{Q}(\sqrt{30})$.

At $c = 13$: the ratio is $87/87 = 1$. Self-duality forces field-preservation.

For the Ising model: the ratio is $49/299 = 49/(13 \cdot 23)$, which is not a perfect square. The Ising gravity at $c = 1/2$ lives over $\mathbb{Q}(\sqrt{5})$; its Koszul dual at $c = 51/2$ lives over $\mathbb{Q}(\sqrt{1495})$ where $1495 = 5 \cdot 13 \cdot 23$. Different fields, different class numbers, different arithmetic. Koszul duality does not preserve the gravitational number field at the Ising point.

This failure is generic. Among all minimal models $M(p, q)$ with $p < 10$, none has the Koszul pair giving the same field. The criterion $(5c+22)/(152-5c) \in \mathbb{Q}^{\times 2}$ requires an arithmetic coincidence that the minimal model spectrum does not provide.

*

What the class number measures. The class number h of $K(c)$ is a measure of the arithmetic complexity of gravity at central charge c. In algebraic number theory, h counts the failure of unique factorisation in the ring of integers $\mathcal{O}_K$.

In the gravitational context: $h = 1$ means the shadow tower's arithmetic is "principal," governed by a single generator. $h > 1$ means the shadow arithmetic requires multiple ideal classes. Whether this has a direct physical interpretation (a counting of distinct gravitational saddle classes, perhaps) is not known.

c	h	**Interpretation**
$1/2$ (Ising)	1	principal: simplest arithmetic
$4/5$ (Potts)	2	one non-trivial ideal class
$7/10$ (tricrit.)	4	four ideal classes
13 (self-dual)	8	most complex in standard range

*

What fails. The growth rate $\rho(c)$ is a smooth algebraic function of c. The fundamental unit $\varepsilon(d(c))$ of $K(c)$ depends on c through the squarefree part $d(c)$, which is a wildly discontinuous function of the central charge. No algebraic relation connects ρ and ε.

A conjecture that the fundamental unit of the gravitational number field controls the shadow growth rate was tested by computing both quantities at $c = 1/2, 7/10, 4/5, 1, 13, 25, 26$ and found to be false. At $c = 1/2$: $\rho \approx 12.53$ while $\varepsilon = (1+\sqrt{5})/2 \approx 1.618$. At $c = 13$: $\rho \approx 0.467$ while $\varepsilon = 59+2\sqrt{870} \approx 117.99$. The two quantities vary in opposite directions.

The failure is structural: $\rho(c)$ is a smooth function of c, while $\varepsilon(d(c))$ is a step function. A smooth function cannot equal a step function.

*

What remains. The specialisation $c \mapsto K(c)$ is a new functor from (a subset of) the rational points of the Virasoro central charge line to the set of real quadratic number fields. It sends the OPE data at genus 0 (the stress tensor and its weight-4 descendant) to a number field whose arithmetic is entirely determined by the bar complex.

The family $\{K(c_{p,q})\}_{(p,q)}$ attached to the minimal model tower constitutes a new arithmetic invariant of 2d CFT. Whether this family has structure beyond what is visible in the table above is an open question.

The first instance of the specialisation functor applied to gravity is the Ising model. The shadow tower of the Ising model lives over $\mathbb{Q}(\sqrt{5})$, the field of the golden ratio. This is the number field with class number 1 and the smallest possible discriminant among real quadratic fields with fundamental unit of norm -1. The simplest unitary minimal model produces the simplest real quadratic field.

Whether this simplicity persists along the minimal model tower, or whether more complex arithmetic emerges, is a question at the intersection of vertex algebra theory and algebraic number theory that the bar construction makes precise.

The bar complex is a 1-form.

Its square root is a number field.

Gravity at rational central charge is arithmetic.

30 Fourier duality from operads to L-functions

The bar construction is a Fourier transform (§**??**, Volume I). The shadow tower is a quadratic extension (§28). The Dirichlet-sewing lift $S_{\mathcal{A}}(u)$ is a product of zeta functions (§**??**, Volume I). These three objects arise from the same chiral algebra. This section asks: do they form a "descent chain" in which a single Fourier duality descends from operadic algebra to analytic number theory?

The answer is: partially. The chain has five levels and three distinct involutions. The first two links are theorems. The third is broken. The fourth is lattice-specific. The claim "they are the same involution" is false.

30.1 The three involutions

The physical content of the bar construction is that it extracts the OPE residues of a chiral algebra into a nilpotent differential, converting the multiplicative structure (operator product) into an additive structure (differential). This extraction carries three symmetries:

(i) **Verdier duality** $\mathbb{D}_{\mathrm{Ran}}$ on $\mathrm{Ran}(X)$. Exchanges $\overline{B}(\mathcal{A})$ and $\overline{B}(\mathcal{A}^!)$ (Theorem A, Volume I). This is the Fourier involution: it intertwines convolution with pointwise operations on factorisation (co)algebras. Its kernel is $\eta_{ij} = d\log(z_i - z_j)$.

Fixed locus: self-dual factorisation coalgebras.

(ii) **Galois involution** σ of the shadow extension $K_L = \mathbb{Q}(c)(t)(\sqrt{Q_L})$. Sends $\sqrt{Q_L} \mapsto -\sqrt{Q_L}$. This is the monodromy of the shadow connection ∇^{sh}: transport around a branch point of $\sqrt{Q_L}$ returns $-\sqrt{Q_L}$. It is the Koszul sign $(-1)^n$ at bar degree n.

Fixed locus: the base field $F = \mathbb{Q}(c)(t)$.

(iii) **Koszul involution** $\mathcal{A} \mapsto \mathcal{A}^!$ on families of chiral algebras. For Virasoro: $c \mapsto 26-c$. For Kac–Moody: $k \mapsto -k - 2h^\vee$. This acts on the *coefficients* of the shadow metric Q_L, not on $\sqrt{Q_L}$ itself.

Fixed locus: self-dual algebras ($c = 13$ for Virasoro).

These three involutions are related but distinct. Verdier duality produces the Koszul involution as its shadow on moduli: $\mathbb{D}_{\mathrm{Ran}}(\overline{B}(\mathcal{A})) \simeq \overline{B}(\mathcal{A}^!)$ implies that $\mathcal{A} \mapsto \mathcal{A}^!$ is the trace of Verdier duality on isomorphism classes. The Galois involution is intrinsic to a *single* algebra's shadow tower and has no direct relation to Verdier duality. It arises because the MC equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$, projected to the scalar arity line, reduces to a Riccati equation whose solution $H(t) = t^2\sqrt{Q_L}$ is algebraic of degree 2—not higher, because the Riccati is quadratic.

The claim "they are the same involution (Fourier) acting at different levels" conflates (i)–(iii). The correct statement: they are three $\mathbb{Z}/2$-actions on three different categories, all originating from the bar construction but not related by any "descent." At the self-dual point $c = 13$, all three have compatible fixed-point behaviour (the Koszul pair coincides, the shadow extensions K_L and $K_{L^!}$ agree, and the Verdier dual coalgebra is isomorphic to the original). At generic c, they diverge: the Galois involution of K_L has nothing to do with the Koszul involution on the c-line.

30.2 Where the chain breaks

The descent chain has five levels:

Level	Object	Status
0	Koszul duality $\mathcal{A} \mapsto \mathcal{A}^!$	Proved (Theorem A)
1	Bar as Fourier: $\overline{B}_X$, inversion, Plancherel	Proved (Theorems A,B,C)
2	Shadow tower as quadratic extension	Proved
3	Dirichlet-sewing lift $S_{\mathcal{A}}(u)$	Defined, but independent of Level 2
4	Zeros of ε_s^c on critical lines	Lattice only; structural obstruction

The break occurs between Levels 2 and 3. The shadow tower is determined by (κ, α, S_4)—the arity-2, 3, 4 OPE data—and lives over $\mathbb{Q}(c)(t)$. The Dirichlet-sewing lift is determined by the weight multiset $W(\mathcal{A}) = \{w_i\}$ and lives over the complex u-plane. These are independent invariants: two algebras with the same weight multiset but different OPE structure constants have the same $S_{\mathcal{A}}(u)$ but different shadow towers. Conversely, two algebras with the same (κ, α, S_4) but different weight multiplicities have the same shadow tower but different $S_{\mathcal{A}}(u)$.

30.3 The structural obstruction

For lattice VOAs, the constrained Epstein zeta ε_s^c factors into L-functions because the theta function Θ_Λ decomposes under the Hecke algebra into Eisenstein series and eigenforms. This decomposition uses three properties special to the lattice setting: the theta function is a classical modular form, the Hecke decomposition is a finite sum, and the Fourier coefficients are lattice representation numbers with multiplicative structure.

None of these holds for the Virasoro algebra. The Virasoro character $q^{-c/24} \prod (1-q^n)^{-1}$ is not a modular form of finite weight. The shadow tower still runs (class M, infinite depth), but the arithmetic sieve does not apply. The MC equation constrains the spectral coefficients on the *real* spectral axis; the zeta zeros live at *complex* spectral parameters. This separation is structural (Remark **??**, Volume I): algebraic constraints on the spectral line cannot reach the scattering poles without analytic continuation of the spectral data off the real axis.

Concretely: the modified Li coefficients $\tilde{\lambda}_n(\mathcal{H})$ for the Heisenberg sewing lift $S_{\mathcal{H}}(u) = \zeta(u)\zeta(u+1)$ are positive for $n = 1, \ldots, 6$ and **negative at** $n = 7$ (`compute/tests/test_euler_koszul_engine.py`, line 394). Positivity of all Li coefficients would be equivalent to GRH for $\zeta(u)\zeta(u+1)$. The sign change at $n = 7$ shows that the sewing-lift Li coefficients do not satisfy a "shadow RH." (This does not contradict RH for ζ alone; the sign change is an artefact of the product and the regularisation.)

30.4 What the bar construction does provide

The descent chain, as a linear sequence, is broken. But the bar construction does provide something more interesting: a *fan* of three independent projections from a single MC element.

Projection	Output	Content
Genus tower	$F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$	Topological (Mumford classes)
Shadow tower	$\{S_r\}_{r\geq 2} = [t^{r-2}]\sqrt{Q_L}$	Arithmetic (quadratic extension, discriminant)
Koszul dual	$\mathcal{A}^! = (\overline{B}(\mathcal{A}))^\vee$	Algebraic (Verdier dual)

These three projections are facets of $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})$, the universal MC element in the modular cyclic deformation complex. The genus tower is the scalar projection of $\Theta_{\mathcal{A}}$ to each genus. The shadow tower is the arity projection of $\Theta_{\mathcal{A}}$ on the primary line. The Koszul dual is the genus-0, arity-2 projection (the binary pairing that determines $\mathcal{A}^!$).

In the gravitational context ($\mathcal{A} = \mathrm{Vir}_c$):

- The genus tower gives $F_g = (c/2)\lambda_g^{\mathrm{FP}}$: the Bernoulli numbers. Higher-genus gravitational partition functions.
- The shadow tower gives $\sqrt{Q_{\mathrm{Vir}}}$: the quadratic extension $K = \mathbb{Q}(c)(t)(\sqrt{Q})$ with discriminant $-320c^2/(5c+22)$. The arithmetic of quantum gravity.
- The Koszul dual gives Vir_{26-c}: the dual gravitational theory. $c + c' = 26$ is Freudenthal–de Vries.

All three are proved. All three are outputs of the single object Θ_{Vir}. But they are three outputs, not five levels of one chain.

30.5 The correct picture

The bar construction is a Fourier transform. This is a theorem, not a metaphor (Volume I, Theorems A, B, C). The shadow tower is a quadratic extension. This is a theorem, not an analogy (§28). But the claim that these two facts are levels of a "descent chain" that continues to L-functions and the Riemann hypothesis is false.

The reason it is false is not a failure of imagination; it is a theorem. The structural obstruction (Remark **??**) shows that algebraic constraints from the MC equation act on the real spectral line, while the zeta zeros live at complex spectral parameters. No amount of clever identification can bridge this gap: it is the difference between a finite-dimensional algebraic structure (the shadow tower, determined by three numbers κ, α, S_4) and an infinite-dimensional analytic object (the zeta function, with infinitely many independent zeros).

The shadow tower *detects* arithmetic—Hecke eigenforms for lattice VOAs, quadratic fields for all VOAs at rational c—but it does not *control* the arithmetic. The detection is a theorem (Theorem **??**). The control would require a mechanism for the MC equation to force zeros onto critical lines, and no such mechanism exists.

What remains is beautiful and proved. The bar construction of a chiral algebra is a Fourier transform that simultaneously produces topological invariants (Mumford classes), arithmetic invariants (quadratic fields), and algebraic invariants (Koszul duals) from a single MC element. The gravitational shadow tower, specialised to rational central charge, assigns a number field to each minimal model. The Ising model lives over $\mathbb{Q}(\sqrt{5})$. The self-dual gravity at $c = 13$ lives over $\mathbb{Q}(\sqrt{870})$. Whether these number fields have structure beyond what is visible in the specialisation table is an open question at the intersection of vertex algebra theory and algebraic number theory.

The bar construction makes the question precise. It does not answer it.

31 Session notes: 1 April 2026

Six findings from the session. The first is a theorem; the remaining five are its consequences.

31.1 The DS-HPL transfer theorem

The single strongest new result.

THEOREM 31.1 *(DS-HPL transfer; proved).* Let $\mathfrak{g}$ be a simple Lie algebra, k a non-critical level, and $\mathcal{W}_k(\mathfrak{g})$ the principal Drinfeld–Sokolov reduction of $V_k(\mathfrak{g})$. The homological perturbation lemma transfers the dg-shifted Yangian datum (m, r, Δ) from the affine side to the $\mathcal{W}$-algebra side via the BRST strong deformation retract (ι, p, h) with $\mathrm{gh}(h) = -1$. The transferred coproduct Δ^W is strictly primitive at every arity:

$$\Delta_z^W(x) = \tau_z(x) \otimes 1 + 1 \otimes x$$

for every $x \in \mathcal{W}_k(\mathfrak{g})$, with no corrections.

Proof. The HPL expansion of the transferred coproduct sums over decorated planar trees. Every tree with at least one internal vertex carries at least one homotopy insertion h, shifting ghost number away from 0. The projection $p \otimes p$ onto ghost-number-0 cohomology annihilates all such trees.

To see that no tree escapes: source trees (the affine coproduct insertions) carry ghost number $+1$, and target trees (the output legs) carry one tensor factor at ghost number -1. Both are killed by $p \otimes p$. The only surviving term is the trivial tree (no internal vertices), which contributes the primitive term.

The argument is independent of the specific Lie algebra, the level, and the arity. It depends only on the ghost number of the BRST homotopy. □

The theorem says that Drinfeld–Sokolov reduction kills the coproduct channel completely. Only the r-matrix channel survives. The $\mathcal{W}$-algebra inherits the A_∞ products m_k^W (infinitely many, for class M) and the collision residue $r^W(z)$, but its coproduct is as trivial as a one-dimensional algebra's. It produces nothing. It only braids.

*

A remark on what is transferred and what is not. The SDR (ι, p, h) transfers three structures simultaneously:

Structure	**Transferred result**	**Status**
A_∞ products m_k	nontrivial for all $k \geq 2$	class M
r-matrix $r(z)$	nontrivial, all poles survive	quasi-trigonometric
Coproduct Δ_z	primitive (all corrections vanish)	proved

The A_∞ products are nontrivial because the perturbation δ_{BRST} has ghost-number-preserving components that survive the projection p. The r-matrix is nontrivial because it arises from the collision residue, which is a genus-0 operation with no homotopy insertions at the output. The coproduct is primitive because it is a genus-0, arity-2, *splitting* operation: the output has two tensor factors, and every homotopy insertion shifts ghost number into the range that $p \otimes p$ annihilates.

The asymmetry is that collision (computing $r(z)$) applies p once, while splitting (computing Δ_z) applies $p \otimes p$: the tensor product doubles the projection's power to kill ghost-shifted outputs.

31.2 Gravitational primitivity as physical principle

Gravity does not produce particles. It only braids them.

In three dimensions, there are no local gravitational degrees of freedom and no propagating gravitons. The bar-cobar machine makes this algebraic: the coproduct channel is trivial, and only the r-matrix channel survives. The gravitational r-matrix

$$r^{\mathrm{Vir}}(z) \;=\; (c/2)/z^3 + 2T/z$$

encodes all gravitational scattering. BTZ black holes are topological: they are monodromies of the r-matrix braiding, not vertices of the coproduct.

*

The (m, Δ)-complexity plane organises the five standard HT theories:

Theory	m-depth	Δ-depth
Chern–Simons	2 (quadratic)	≥ 1 (nontrivial)
Twisted holography	2	≥ 1
M2	2	≥ 1
Gravity	∞ (class M)	0 (primitive)
M5	∞	0

Gravity sits at $(\infty, 0)$: maximal A_∞ complexity, vanishing coproduct complexity. Gauge theory sits at $(2, \geq 1)$: minimal A_∞ complexity, nontrivial coproduct. DS reduction moves a theory from the gauge corner to the gravity corner by killing the coproduct and inflating the A_∞ tower.

31.3 Five worked examples as Θ^{oc} instantiations

The completed eight-fold platonic datum $\Pi_X^{\mathrm{oc}}(\mathcal{A})$ for the five canonical HT systems:

1. Chern–Simons. $\mathcal{A} = V_k(\mathfrak{g})$. Class L, shadow depth 3. $\kappa + \kappa^! = 0$ (Feigin–Frenkel). Coproduct: nontrivial (Casimir cross-term). Sewing: determinant. The entire A_∞ structure is formal: $m_k = 0$ for $k \geq 3$. The platonic package is finite-dimensional at each arity.

2. Three-dimensional gravity. $\mathcal{A} = \mathrm{Vir}_c$. Class M, shadow depth ∞. $\kappa + \kappa^! = 13 \neq 0$. Coproduct: **primitive** (ghost-number obstruction from DS). Sewing: Pfaffian-squared. The quartic contact invariant $Q^{\mathrm{ct}} = 10/[c(5c + 22)]$ controls the entire infinite tower. The r-matrix $r(z) = (c/2)/z^3 + 2T/z$ carries all scattering.

3. Twisted holography. $\mathcal{A} = V_k(\mathfrak{g})$ (affine Kac–Moody, Costello–Gaiotto). Class L, shadow depth 3. Multi-channel: for $\mathcal{W}_3$, the genus-2 cross-channel contribution is

$$F_2^{\mathrm{cross}} = \frac{c + 120}{16c}.$$

The holographic dictionary is the Koszul triangle read through Theorems A–H.

4. M2-brane. $\mathcal{A} = \mathrm{DDCA}_k(\mathfrak{gl}_K)$. Class L. Coproduct: nontrivial. Two-variable r-matrix $r(z_1, z_2)$. No DS reduction: the $\mathcal{W}$-algebra structure is absent. This is gauge theory, not gravity, and the nontrivial coproduct confirms it.

5. M5-brane / 6d $(2, 0)$. $\mathcal{A} = \mathcal{W}_\infty(K)$ at large N. Class M, shadow depth ∞. Coproduct: **primitive** (same ghost-number argument as Virasoro, since $\mathcal{W}_N$ arises by DS reduction). Per-spin r-matrix. Shadow depth increases under DS.

*

The shadow dichotomy across the five theories:

Lineage	**Class**	**Mechanism**
Current-algebra (CS, M2)	L	finite towers, no DS
$\mathcal{W}$-algebra (gravity, $\mathcal{W}_3$, M5)	M	infinite towers, DS origin

DS is the shadow-depth escalator. It converts class L (finite, formal) to class M (infinite, genuinely A_∞). The ghost-number filtration that kills the coproduct is the same filtration that inflates the A_∞ tower. The two effects are a single mechanism.

31.4 The open-sector primitive

The primitive object is $\mathcal{C}_{\mathrm{op}}$, not $\mathcal{A}$ or $\overline{B}(\mathcal{A})$.

For Chern–Simons: $\mathcal{C}_{\mathrm{op}} = V_k(\mathfrak{g})$-mod, the category of $V_k(\mathfrak{g})$-modules as a factorisation dg-category with meromorphic tensor product. The boundary algebra $A_b = \mathrm{End}(\mathrm{trivial})$ recovers $V_k(\mathfrak{g})$. The chiral derived center $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}} = C^\bullet_{\mathrm{ch}}(\mathcal{A}_b, \mathcal{A}_b)$ classifies bulk operators.

Modularity is trace plus clutching on $\mathcal{C}_{\mathrm{op}}$, not an axiom on the closed algebra. The annulus trace

$$\mathrm{Tr}\colon K_0(\mathcal{C}_{\mathrm{op}}) \longrightarrow Z_1(\mathcal{A})$$

gives the genus-1 shadow. The full genus-g amplitude is the composition of g clutching maps on $\mathcal{C}_{\mathrm{op}}$. The modular S-matrix is a derived property of the trace, not an input to the theory.

The boundary algebra A_b is a chart: a Morita-dependent presentation of $\mathcal{C}_{\mathrm{op}}$. Different compact generators produce Morita-equivalent charts. The category is the invariant; the algebra is a coordinate system.

31.5 BV/BRST = bar: honest status

Three levels of identification. The first is proved; the second is conditionally proved; the third is conjectural.

Genus	Claim	Status
$g = 0$	algebraic BRST = bar differential	proved
$g = 0$, higher arity	Feynman-diagrammatic m_k = bar m_k	proved
$g \geq 1$	BV quantum master equation = curved bar	conjectural

The bar complex IS well-defined at all genera: the inductive genus determination from Volume I combined with the HS-sewing theorem produces bar amplitudes at every genus. What is open is whether these bar amplitudes coincide with the BV-BRST partition function at genus $g \geq 1$. This is `conj:master-bv-brst`.

The identification at genus 0 is an algebraic comparison: the BRST differential $Q = \partial + \partial^{\mathrm{BRST}}$, restricted to the physical subspace, reproduces the bar differential $d_{\overline{B}}$. The identification at genus 1 would require matching the BV quantum master equation $\hbar\Delta_{\mathrm{BV}}S + \frac{1}{2}\{S, S\} = 0$ with the curved bar relation $d^2_{\mathrm{fib}} = \kappa \cdot \omega_1$. No obstruction to this matching is known; no proof exists.

This session corrected four locations in the manuscript that overclaimed genus-1 BV/BRST = bar as proved. The bar complex is the proved object. The BV-BRST comparison is the conjectural bridge.

31.6 The standalone paper

Shadow Towers and Algebraicity, 25 pages, eight sections.

- §1. Introduction and motivation. The bar construction as categorical logarithm; the shadow tower as its Taylor expansion.
- §2. The MC algebra. The modular cyclic deformation complex $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A})$ and the universal element $\Theta_{\mathcal{A}}$.
- §3. The shadow tower. Projection to the primary line; recursive computation of S_r from the MC equation.
- §4. The Riccati algebraicity theorem. The MC equation on a single primary line reduces to a Riccati ODE whose solution is $H(t) = t^2\sqrt{Q_L(t)}$, algebraic of degree 2. Full proof.

§5. Four-class classification. The critical discriminant $\Delta = 8\kappa S_4$ partitions modular Koszul algebras into four classes:
   (a) **G** (Gaussian): $r_{\max} = 2$. Heisenberg.
   (b) **L** (Lie/tree): $r_{\max} = 3$. Affine Kac–Moody.
   (c) **C** (contact/quartic): $r_{\max} = 4$. $\beta\gamma$.
   (d) **M** (mixed): $r_{\max} = \infty$. Virasoro, $\mathcal{W}_N$.

§6. Explicit shadow towers. Complete computation for the standard landscape: Heisenberg, affine $\mathfrak{sl}_2$, $\beta\gamma$, Virasoro, $\mathcal{W}_3$. Tables of S_r through arity 8.

§7. $\mathcal{W}_3$ cross-channel genus-2 computation. Multi-generator shadow tower with two primary lines (T and W), propagator variance δ_{mix}, genus-2 cross-channel formula.

§8. Outlook. Pixton conjecture (class M algebras generate the Pixton ideal via infinite MC tower), analytic completion programme, non-standard families.

The paper extracts from Volume I the self-contained story of the shadow tower, from its definition through the algebraicity theorem to the four-class classification and the explicit standard-landscape computations. It is the first accessible entry point to the monograph's nonlinear characteristic layer.

32 The graviton is composite

The Sugawara stress tensor

$$T \;=\; \frac{1}{2(k+h^\vee)} \sum_a :J^a J_a:$$

is quadratic in affine currents. It is not a fundamental field. The graviton is composite.

The BRST complex of the Drinfeld–Sokolov reduction remembers this. The BRST homotopy h maps states at ghost number 0 to states at ghost number ± 1. Every tree in the homological perturbation lemma expansion of the transferred coproduct passes through h. The projection $p \otimes p$ onto ghost-number-0 cohomology annihilates both signs. The transferred coproduct is

$$\Delta^W(x) \;=\; \tau_z(x) \otimes 1 \;+\; 1 \otimes x.$$

Primitive. No cross-term. No splitting. At every arity, at every order, for every principal DS reduction of every simple Lie algebra.

*

In gauge theory: the gluon J^a IS the connection. Fundamental. Ghost number 0. No passage through h. The BRST homotopy does not obstruct the transferred coproduct. The Casimir cross-term survives projection:

$$\Delta_z(T_{\mathrm{aff}}) \;=\; \tau_z(T_{\mathrm{aff}}) \otimes 1 \;+\; 1 \otimes T_{\mathrm{aff}} \;+\; \frac{1}{k+h^\vee}\,\Omega(w{+}z,\, w),$$

where $\Omega = \sum_a J^a \otimes J_a$ is the split Casimir. A production vertex. A gluon bound state dissociates.

In gravity: the graviton T IS the curvature. Composite. Ghost-burdened. Every HPL tree at ghost number 0 dies under projection. Gravitons do not split. They braid.

*

The r-matrix $r^{\mathrm{Vir}}(z) = (c/2)/z^3 + 2T/z$ carries all gravitational scattering. The cubic pole: the bar kernel $d\log(z-w)$ absorbs one power from the quartic OPE. The Virasoro OPE has poles at z^{-4}, z^{-2}, z^{-1}. The r-matrix has poles at z^{-3}, z^{-1}. The logarithmic derivative is always one degree milder than the function itself. This absorption is not an approximation. It is the structural content of calling the bar construction a categorical logarithm.

*

Five theories. One machine.

	CS	Gravity	TH	M2	M5
Class	L	M	L	L	M
Δ_L	0	$\neq 0$	0	0	$\neq 0$
Coproduct	nontrivial	primitive	nontrivial	nontrivial	primitive
Why	fundamental	ghost kills	fundamental	DDCA, no DS	ghost kills
$r_{\max}$	3	∞	3	3	∞

The pattern: DS reduction creates depth (L $\to$ M) and kills the coproduct. The Sugawara construction is the algebraic mechanism. The ghost sector is the memory of composition. Gauge theories have fundamental fields and nontrivial coproducts. Gravitational theories have composite fields and primitive coproducts. The distinction is not parametric (it is not the value of the central charge). It is structural (it is the ghost-number filtration of the BRST complex).

*

The bar complex of a chiral algebra knows more about quantum gravity than any perturbative expansion.

It knows the exact genus-g free energy at every genus, in closed form: $F_g = (c/2) \cdot \lambda_g^{\mathrm{FP}}$, the $\hat{A}$-genus coefficients times the modular characteristic.

It knows the entire infinite tower of higher-arity corrections, as the Taylor expansion of an algebraic function of degree 2: $H(t) = t^2\sqrt{Q_{\mathrm{Vir}}(t)}$, three numbers (κ, α, S_4), one quadratic form, one square root.

It knows that the gravitational coproduct is primitive: black holes do not split. The production vertex that would allow a graviton to dissociate into constituents is killed by the ghost-number projection at every arity, for every principal Drinfeld–Sokolov reduction.

It knows the arithmetic. At rational central charge c_0, the shadow metric Q_{Vir} specialises to a specific quadratic polynomial over $\mathbb{Q}$, and $K(c_0) = \mathbb{Q}(\sqrt{d(c_0)})$ is a specific imaginary quadratic number field. The Bekenstein–Hawking entropy $S = 2\pi\sqrt{ch/6}$ is the leading coefficient. The discriminant $-320c^2/(5c+22)$ classifies the depth. The class number of $K(c_0)$ measures the arithmetic complexity of gravity at that central charge. The Ising model lives over $\mathbb{Q}(\sqrt{-10})$. The self-dual gravity at $c = 13$ lives over $\mathbb{Q}(\sqrt{-870})$, and the complementarity constant $6960 = 2^4 \cdot 3 \cdot 5 \cdot 29$ appears as $-80 \cdot 87 = -6960$.

*

The bar complex of a chiral algebra is a 1-form, a square root, and a ghost.

The 1-form $\eta = d\log(z_i - z_j)$ extracts the OPE. Arnold's relation makes it nilpotent. The genus-completed differential $D_{\mathcal{A}}$ satisfies $D_{\mathcal{A}}^2 = 0$, and $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0$ is the universal MC element.

The square root $\sqrt{Q_L}$ generates the tower. One quadratic form, one algebraic function, infinitely many rational invariants. The depth classification, the growth rate, the convergence behaviour, the Galois theory, the Epstein zeta function, the discriminant complementarity: all are properties of this single square root.

The ghost kills the splitting. The BRST homotopy carries ghost number ± 1, the projection annihilates it, and the coproduct vanishes. This is the algebraic reason that gravity does not produce particles.

Everything else is a projection of these three.

33 Koszul–Epstein and quantum gravity

33.1 The arithmetic content of the gravitational bar complex

The bar complex of the Virasoro algebra at central charge c carries three layers of arithmetic data.

Layer 1: the Heisenberg core. The Miura map $\hat{\Upsilon}\colon W_\kappa(\mathfrak{sl}_2) \to V^{\mathfrak{h}_\kappa}(\mathfrak{h})$ embeds the Virasoro algebra into a rank-1 Heisenberg. At the level of the sewing determinant, this produces an exact Euler product:

$$\sum_{N\geq 1} \sigma_{-1}(N)\, q^N \;=\; -\log\bigl(\eta(q)^2\bigr) - \tfrac{\tau}{12}, \qquad \sum_{N\geq 1} \sigma_{-1}(N)\, N^{-s} \;=\; \zeta(s)\,\zeta(s+1).$$

The Heisenberg core is arithmetically transparent: all nontrivial genus-one arithmetic is a consequence of the divisor function σ_{-1} being multiplicative.

Layer 2: the Miura defect. The vacuum character of W_N splits as $\chi^{\mathrm{vac}}_{W_N}(q) = \chi_H(q)^{N-1} \cdot \prod_{m=1}^{N-1}(1 - q^m)^{N-m}$. All nontrivial genus-one arithmetic beyond the Heisenberg core sits in the finite Miura defect factor $D_N(q) = \prod_{m=1}^{N-1}(1-q^m)^{N-m}$. For $N = 2$ (Virasoro): $D_2(q) = 1 - q$, removing the weight-1 mode.

Layer 3: the shadow tower. The infinite shadow tower $\{S_r(c)\}_{r\geq 2}$, determined by (κ, α, S_4) via the MC recursion, encodes the full A_∞ deformation data. Its spectral measure ρ is uniquely determined by the Carleman condition. The MC recursion propagates Hecke equivariance from the character level to all arities (Route C, proved for the standard landscape).

33.2 The gravitational dichotomy and the Koszul–Epstein frontier

The gravity dichotomy at $c = 26$ concerns the vanishing of $\kappa_{\mathrm{eff}} = \kappa(\mathrm{matter}) + \kappa(\mathrm{ghost})$. The Koszul–Epstein question concerns the critical-line properties of the arithmetic shadow. These are unrelated:

- $\kappa_{\mathrm{eff}} = 0$ at $c = 26$ determines whether the genus tower terminates (the Clifford-to-exterior degeneration of the transgression algebra).
- The critical-line question concerns the zeros of $\zeta(s)$ appearing in the Heisenberg sewing determinant. These zeros are independent of the central charge: the sewing lift $S_{\mathrm{Vir}}(u) = \zeta(u+1)(\zeta(u) - 1)$ is the same function for all c.

The Koszul self-duality point $c = 13$, where $\kappa + \kappa' = 13$ and $\delta_\kappa = 0$, is the point where Verdier–Hecke commutation forces the constrained Epstein zeta to factor into standard L-functions. This is a genuine arithmetic structure of the self-dual gravitational theory. It does not, however, produce a critical-line theorem.

33.3 Honest assessment

The MC element $\Theta_{\mathcal{A}}$ of the gravitational bar complex produces genuine arithmetic structure: the Carleman-determined spectral measure, Route C prime-locality, the shadow–spectral correspondence for lattice VOAs, the Verdier–Hecke factorisation at self-dual points. These are proved theorems.

The programme does not produce a critical-line theorem. The structural obstruction is that the shadow spectral measure and the automorphic spectral measure are different objects, and the map between them (the Rankin–Selberg integral) is not sign-preserving. No mechanism for positivity has been identified. The bar–cobar homotopy equivalence provides deformation-theoretic transport, not arithmetic positivity.

The surviving content: the quartic resonance class $R_4^{\mathrm{mod}}(\mathcal{A})$ is the first nonlinear place where Koszul duality constrains the automorphic spectral decomposition beyond the character level. Whether this constraint carries critical-line information is open. The correct next computation is the quartic Gram-side matching with the crossing-weighted bilinear residue kernel (using actual DOZZ/Toda structure constants), not the bare spectral measure (which fails by four to five orders of magnitude).

The bar complex of a chiral algebra knows the exact genus tower, the arithmetic of the shadow, and the depth classification. It does not know the zeros of $\zeta(s)$.

34 Spectrum and gravity

The gravitational bar complex carries three kinds of data: the algebraic engine (Theorems A–H, the MC element $\Theta_{\mathcal{A}}$), the arithmetic shadow (the shadow tower, the depth classification, the Epstein zeta), and the spectral content (the primary spectrum, the partition function, the constrained Epstein series ε_s^c). These three overlap but do not coincide. This section records the definitive assessment of their relationship in the gravitational context.

34.1 The spectrum as a categorical invariant

The scalar primary spectrum $\mathrm{Spec}_{\mathrm{sc}}(\mathcal{A}) = \{(\Delta_i, m_i)\}$ of a 2d CFT at central charge c is an invariant of the *modular tensor category* $\mathcal{A}\text{-mod}^{\mathrm{ss}}$ together with the modular-invariant partition function $Z(\tau,\bar{\tau}) = \sum_{i,j} M_{ij}\,\chi_i(\tau)\,\overline{\chi_j(\tau)}$. The multiplicity matrix M_{ij} encodes the physical spectrum; the characters χ_i encode the algebraic spectrum.

The bar complex $B(\mathcal{A})$ is built from the algebra $\mathcal{A}$ (specifically, from the vacuum module and its OPE), not from the category of modules. The bar-cobar machine determines the Koszul dual $\mathcal{A}^!$, the MC element $\Theta_{\mathcal{A}}$, and the shadow tower $\{S_r\}$. It does *not* determine the modular S-matrix or the multiplicity matrix. The primary spectrum is therefore external to the bar construction.

For the gravitational theory (Virasoro at central charge c), the vacuum module is the unique irreducible highest-weight module $L(c,0)$, and the partition function $Z_{\mathrm{grav}} = |\chi_0(\tau)|^2 + \sum_{h>0} m_h\,|\chi_h(\tau)|^2$ depends on the dynamical data $\{m_h\}_{h>0}$: the spectrum of primary states. The bar complex of Vir_c sees the algebra, not the specific gravity theory built from it.

34.2 Which constructions partially capture the spectrum

Five constructions each capture a shadow of the gravitational primary spectrum. None captures it fully from within the bar machine.

1. *Zhu's algebra* $A(\mathrm{Vir}_c)$. For the universal Virasoro: $A(V_c) \cong \mathbb{C}[x]$. The simple modules of $\mathbb{C}[x]$ correspond to points $x = h \in \mathbb{C}$: the conformal dimensions. Zhu's algebra determines the *set* of allowed lowest weights $\{h_{r,s}\}$ for the minimal model at $c = c_{p,q}$ (via the quotient $A(L(c,0)) = \mathbb{C}[x]/(f_c(x))$).

But it does not determine the partition function multiplicities. At generic c (irrational, or rational non-minimal), $A(V_c) = \mathbb{C}[x]$ admits a continuum of simple modules, and no finite selection rule from $A(V_c)$ picks out the physical spectrum.

2. *Associated variety* X_{Vir_c}. For Virasoro: $X_{V_c} = \mathbb{C}$ (one-dimensional, since Virasoro has a single strong generator). The dimension of X_V controls the asymptotic growth of the character and hence the leading pole of the constrained Epstein series at $s = c/2$. This is the Cardy regime: $\log \rho(E) \sim 2\pi\sqrt{c\,E/3}$ for large conformal dimension E.

3. *Bar cohomology* $H^*(B(\mathrm{Vir}_c))$. Concentrated in bar degree 1 (Koszulness). The graded dimensions $\dim H^1(B(\mathrm{Vir}))_h$ count generators of the Koszul dual $\mathrm{Vir}^!_{26-c}$ at weight h. These are the *Motzkin differences* $M(h+1) - M(h)$, supported only at even weights, independent of c, and growing as 3^h. The quasi-primary count (partitions minus descendants) is a different sequence, nonzero at odd weights and growing as $e^{\pi\sqrt{2h/3}}$. Bar cohomology sees the dual algebra, not the spectrum.

4. *Shadow tower* $\{S_r\}$. The shadow tower extracts three numbers (κ, α, S_4) from the OPE and packages them into the shadow metric Q_L. The shadow-metric Epstein zeta $\varepsilon_{Q_L}(s)$ is a binary quadratic form sum, unrelated to the primary spectrum except through a chain of projections. For lattice VOAs, the shadow–spectral correspondence recovers the full L-function package; for Virasoro, the correspondence is incomplete (the shadow has 3 parameters; the spectrum has infinitely many).

5. *The chiral derived centre* $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$. The universal bulk algebra, built as chiral Hochschild cochains $C^\bullet_{\mathrm{ch}}(\mathcal{A}_b, \mathcal{A}_b)$, classifies bulk operators. Its cohomology controls the BRST-closed observables of the 3d HT theory. But it is a genus-0 invariant of the open/closed Swiss-cheese structure, and does not directly encode the genus-1 partition function. The trace on the derived centre (the annulus trace $\mathrm{Tr}_{\mathcal{A}}$) recovers the Hochschild homology $\mathrm{HH}_*(\mathcal{A})$, which is the first modular shadow of the open sector. This is related to, but not identical with, the partition function.

34.3 The gravitational spectral gap

In the gravitational context, the primary spectrum has additional structure: the conformal bootstrap imposes crossing-symmetry constraints on the four-point function, which restrict the allowed $\{(\Delta_i, m_i)\}$ beyond what the chiral algebra alone determines. The Benjamin–Chang constrained Epstein series ε^c_s is the Eisenstein spectral coefficient of $\hat{Z}^c$, and its functional equation involves $\zeta(2s)/\zeta(2s-1)$, introducing poles at $s = (1+\rho_n)/2$ where ρ_n are nontrivial zeros of ζ.

The pole *locations* are universal: they come from $\zeta(2s-1) = 0$ in the Γ-factor ratio $F_c(s)$, and do not depend on the specific CFT. The pole *residues* are CFT-specific: they are integrals of the primary-counting function against residues of Eisenstein series on the modular fundamental domain. The bar complex constrains neither the locations (which are number-theoretic: the zeros of ζ) nor the residues (which are spectral: the primary multiplicities m_h).

For the gravitational theory at $c = 26$ (critical string): $\kappa_{\mathrm{eff}} = \kappa(\mathrm{matter}) + \kappa(\mathrm{ghost}) = 0$, and the genus tower terminates. But the primary spectrum is still infinite (the Virasoro module at $c = 26$ has infinitely many primaries at all weights $h \geq 2$), and ε^{26}_s is a nontrivial Dirichlet series. The genus tower's termination is a statement about the $\hat{A}$-genus factor κ; it says nothing about the spectral content.

For the gravitational theory at $c = 13$ (self-dual point): Verdier–Hecke commutation forces the constrained Epstein to factor into standard L-functions. This is the strongest structural result: self-duality of the chiral algebra produces a factorisation of the spectral invariant. But even at $c = 13$, the factorisation does not determine the zeros of the constituent L-functions.

34.4 Is ε_s^c a bar-complex invariant?

PROPOSITION 34.1 *(The constrained Epstein is not a bar-complex invariant).* The constrained Epstein series $\varepsilon_s^c(\mathcal{A})$ cannot be reconstructed from the bar complex $B(\mathcal{A})$ and its cohomology.

Proof. Three independent arguments.

(a) Categorical argument. Two non-isomorphic modular-invariant partition functions for the same chiral algebra produce different constrained Epstein series. Example: the Ising model ($c = 1/2$) has the diagonal partition function $Z_{\mathrm{diag}} = |\chi_0|^2 + |\chi_{1/2}|^2 + |\chi_{1/16}|^2$ and the charge-conjugation partition function (which is the same, since Ising is self-conjugate). But for $\widehat{\mathfrak{su}(2)}_k$ at $k \geq 10$, there exist exceptional modular invariants (the D-type and E-type classifications of Cappelli–Itzykson–Zuber) that produce different primary spectra from the same chiral algebra. The bar complex $B(\hat{\mathfrak{g}}_k)$ is the same for all choices of modular invariant; the constrained Epstein series differs.

(b) Dimensional argument. The bar complex is determined by the OPE data of the vacuum module (a countable set of structure constants). The primary spectrum is determined by the modular invariant *and* the full representation theory (the braided tensor structure, the modular S-matrix). The latter is strictly richer: two algebras with the same OPE but different categories of modules (e.g., $V_k(\mathfrak{g})$ at the same k but with different conformal extensions) have the same bar complex but different spectra.

(c) Weight argument. $\dim H^1(B(\mathrm{Vir}))_h$ at odd h vanishes (the Koszul dual has generators only at even weights), while the quasi-primary count at odd h is generically nonzero. No Dirichlet series built from bar cohomology dimensions can reproduce ε_s^c, which sums over all primary weights including odd ones. □

34.5 The honest three-layer picture

The gravitational bar complex produces three layers of arithmetic data, each genuine and proved.

Layer 1: the genus expansion. $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}}$ for uniform-weight algebras. This is the leading scalar projection of the MC element $\Theta_{\mathcal{A}}$, generated by $\hat{A}(i\hbar) - 1$. It captures *one number* (κ) from the full partition function.

Layer 2: the shadow tower. $\{S_r\}_{r\geq 2}$, determined by (κ, α, S_4). This captures *three numbers* from the OPE, packaged into the shadow metric Q_L and the quadratic extension K_L. The shadow–spectral correspondence for lattice VOAs, Route C prime-locality, and the Verdier–Hecke factorisation are genuine arithmetic results at this layer.

Layer 3: the spectral content. $\varepsilon_s^c(\mathcal{A}) = \sum(2\Delta)^{-s}$, encoding the full scalar primary spectrum. This layer is external to the bar complex: it requires the partition function, which is a genus-1 categorical invariant, not a genus-0 algebraic one.

The gap between Layers 2 and 3 is the gap between the shadow tower (genus-0, 3 parameters) and the primary spectrum (genus-1, ∞ parameters). No mechanism in the MC equation or the bar-cobar machine bridges this gap: the shadow tower is determined by finitely many OPE coefficients, while the spectrum encodes infinitely many independent quantities.

The honest statement for the gravitational theory: *the bar complex of the Virasoro algebra at central charge c determines the exact genus expansion $F_g = (c/2)\,\lambda_g^{\mathrm{FP}}$, the infinite shadow tower with growth rate $\rho(c)$ and discriminant $\Delta = 80/(5c+22)$, and the four-class depth assignment (class M for all $c \neq 0$). It does not determine the primary spectrum, the partition function, or the constrained Epstein series. The spectrum is a representation-theoretic invariant; the bar complex is an algebra-theoretic invariant. They live in different categories.*

35 Derived global sections in quantum gravity

The gravitational bar complex defines a sheaf on moduli space whose derived global sections are algebraic invariants. This section records the definitive assessment of what these algebraic invariants capture in the gravitational context, correcting the false claim that "the bar complex cannot see the spectrum because integration over moduli space is analytic."

35.1 The sheaf-theoretic framework

For Virasoro at central charge c, the bar complex at genus g defines a sheaf of dg modules on $\overline{\mathcal{M}}_g$. The key objects:

1. The *virtual bar K-class* $V_{\mathrm{Vir}_c} := [R\pi_{g*}\overline{B}^{(g)}(\mathrm{Vir}_c)] \in K^0(\overline{\mathcal{M}}_g)$, computed by derived pushforward along the universal curve. This is well-defined: the perfectness criterion (Koszul acyclicity + finite-dimensional fiber cohomology) is satisfied for all modular Koszul algebras.

2. The *center sheaf* $\mathcal{Z}_{\mathrm{Vir}_c} := R^*\pi_{g*}(\overline{B}^{(g)}, d_{\mathrm{fib}})$, a local system on $\overline{\mathcal{M}}_g$ with fiber $H_*^{\mathrm{ch}}(\Sigma_g, \mathrm{Vir}_c)$.

3. The *ambient complex* $C_g(\mathrm{Vir}_c) = R\Gamma(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathrm{Vir}_c})$, the derived global sections. This is a purely algebraic operation on a constructible sheaf over a proper DM stack.

4. The *GRR hierarchy*:

$$\begin{aligned} c_1(\det V_{\mathrm{Vir}_c}) &= \tfrac{c}{2}\cdot\lambda, \\ \mathrm{ch}(V_{\mathrm{Vir}_c}) &\leftrightarrow \Delta_{\mathrm{Vir}}(x) \quad \text{(Motzkin differences)}, \\ \mathrm{Hol}(V_{\mathrm{Vir}_c}) &= \Pi_{\mathrm{Vir}_c} \quad \text{(MCG monodromy)}. \end{aligned}$$

   All three levels are algebraic.

So the bar complex *does* produce algebraic invariants from integration over moduli space. The claim that this integration is "analytic, not algebraic" was a false idea. Derived global sections, Chern character, GRR pushforward, and monodromy representation are all algebraic operations.

35.2 The gravitational spectral gap: precise diagnosis

The GRR hierarchy captures the *algebra's* modular structure. It does not capture the *gravitational spectrum*. The precise diagnosis:

(i) The GRR scalar level recovers $\kappa(\mathrm{Vir}_c) = c/2$. This is one number. The gravitational partition function $Z(\tau,\bar\tau) = |\chi_0(\tau)|^2 + \sum_{h>0} m_h\,|\chi_h(\tau)|^2$ encodes infinitely many numbers $\{m_h\}$.

(ii) The GRR spectral level recovers the bar cohomology dimensions $\dim H^1(B(\mathrm{Vir}))_h$, which are the Motzkin differences $M(h+1) - M(h)$. These count generators of the Koszul dual $\mathrm{Vir}^!_{26-c}$, not primaries of Vir_c. The two sequences differ: bar cohomology is supported only at even weights and is c-independent; the quasi-primary count is nonzero at odd weights and depends on the specific gravity theory.

(iii) The GRR holonomy level recovers the monodromy of $\mathcal{Z}_{\mathrm{Vir}_c}$: the action of the mapping class group on chiral homology. For the Virasoro algebra, this is the monodromy of the KZ connection, which encodes the projective representation of $\mathrm{MCG}(\Sigma_g)$ on the space of conformal blocks. This is an invariant of the *algebra* (it is functorial in morphisms of chiral algebras, not in changes of the module category or modular invariant).

The obstruction is not analysis vs. algebra. It is *algebra vs. category*: the bar complex sees the chiral algebra Vir_c; the gravitational spectrum sees the choice of gravity theory built from Vir_c (the specific multiplicities $\{m_h\}$ in the partition function, which are dynamical data external to the algebra).

35.3 Three routes toward the spectrum

(a) Module bar construction. The module bar complex $\overline{B}_{\mathrm{Vir}_c}(L(c,h))$ applied to each irreducible module $L(c,h)$ produces a comodule over $\overline{B}(\mathrm{Vir}_c)$. Its K-class carries the conformal dimension h. Summing with the gravitational multiplicities $\{m_h\}$:

$$\sum_h m_h \,\mathrm{ch}\big(R\pi_{1*}\,\overline{B}_{\mathrm{Vir}_c}(L(c,h))\big)$$

would recover the gravitational partition function at genus 1. But this takes $\{m_h\}$ as *input*: it computes the partition function from the spectrum, rather than extracting the spectrum from the bar complex.

(b) Conformal blocks as D-module. The conformal blocks local system on $\mathcal{M}_{g,n}$ carries a flat connection (the KZ/Hitchin connection) whose monodromy encodes the modular S-matrix for rational theories. At integrable level for affine Kac–Moody algebras, the monodromy recovers the primary dimensions via the Verlinde formula.

For the Virasoro algebra at generic c, the conformal blocks local system has irregular singularities and the representation theory is non-semisimple (at irrational c, all Verma modules are irreducible and the spectrum is continuous). Whether the D-module structure of the conformal blocks sheaf on $\overline{\mathcal{M}}_{1,1}$ determines the gravitational spectrum (after choosing a modular invariant) is a frontier question: it amounts to recovering the spectral decomposition of a function on $\mathrm{SL}(2,\mathbb{Z})\backslash\mathfrak{H}$ from the algebraic structure of its associated D-module.

(c) The Roelcke–Selberg step. The passage from the partition function $Z(\tau,\bar{\tau})$ to the constrained Epstein series ε_s^c uses the spectral decomposition of $L^2(\mathrm{SL}(2,\mathbb{Z})\backslash\mathfrak{H})$ into Eisenstein series and Maass cusp forms. This involves the hyperbolic Laplacian, which is a real-analytic operator. The scattering matrix involves $\zeta(2s)/\zeta(2s-1)$, whose poles are the nontrivial zeros of the Riemann zeta function. No categorical or motivic construction is known to produce this spectral decomposition.

The "analytic" step that is genuinely unavoidable is this last one: from characters (which are algebraic for rational VOAs) to spectral coefficients (which require harmonic analysis on $\mathrm{SL}(2,\mathbb{Z})\backslash\mathfrak{H}$).

35.4 Honest status

(a) The bar complex defines a sheaf on $\overline{\mathcal{M}}_g$ with algebraic invariants ($R\Gamma$, ch, Hol, GRR). The claim "integration is analytic" was a false idea.

(b) These algebraic invariants capture the algebra's modular structure: κ, bar cohomology dimensions, KZ/Hitchin monodromy. They do *not* capture the gravitational spectrum $\{m_h\}$ or the constrained Epstein ε_s^c.

(c) The obstruction is *algebra vs. category*, not analysis vs. algebra. Two gravity theories with the same Virasoro algebra but different spectra have the same bar complex. This is a theorem, not a technical limitation.

(d) The construction that does capture the spectrum—the conformal blocks D-module on $\mathcal{M}_{g,n}$—is algebraic (Beilinson–Drinfeld) but requires the module category as input. For rational theories, its monodromy (S-matrix) determines the primary dimensions. For non-rational theories, the D-module has irregular singularities and the spectral question is at the frontier of geometric Langlands and irregular Hodge theory.

(e) The single genuinely analytic step is the Roelcke–Selberg decomposition: from characters to spectral zeta. This step is universal (it involves the Riemann zeta function, not the specific VOA) and has no known categorical substitute.

(f) Status: the "spectral question" is **OPEN** in the sense that the algebraic D-module framework exists but has not been carried through for Virasoro at generic c. The gap is not a missing construction but a missing computation: does the irregular D-module structure of the Virasoro conformal blocks local system on $\overline{\mathcal{M}}_{1,1}$ determine the primary spectrum after choosing a modular invariant?

36 Complex parameters in quantum gravity

The gravitational bar complex of the Virasoro algebra produces real spectral data: the modular characteristic $\kappa = c/2$, the shadow tower $\{S_r(c)\}$, and the genus expansion $F_g = \kappa\lambda_g^{\mathrm{FP}}$ are all rational functions of the central charge c. The zeros of the Riemann zeta function, which appear in the scattering matrix of the Eisenstein spectral decomposition on $\mathcal{M}_{1,1}$, live at complex spectral parameters. This section records the definitive assessment of whether any extension of the bar complex can access complex spectral data.

36.1 The structural obstruction

The obstruction is a theorem with three components.

1. The shadow tower on a single primary line is determined by three genus-0 OPE invariants (κ, α, S_4) via Theorem **??**: the generating function $H(t) = t^2\sqrt{Q_L(t)}$ is algebraic of degree 2. Three numbers constitute a finite-dimensional algebraic datum.

2. The zeta zeros $\{\rho_n = \frac{1}{2} + i\gamma_n\}$ constitute a countably infinite collection of complex numbers, believed to be algebraically independent. No finite-dimensional algebraic structure determines such a set.

3. The Rankin–Selberg unfolding that connects the partition function to the constrained Epstein ε_s^c erases the convolution-algebraic structure of the MC equation. The unfolded integral depends on Fourier coefficients alone, not on the MC brackets. The L-functions $L(s, f_j)$ in the spectral decomposition are properties of the eigenforms f_j, independent of the algebra $\mathcal{A}$.

36.2 Eight candidate bridges: the census

A complete investigation examined eight approaches to bridging the gap from real bar-complex data to complex L-function zeros.

Dead ends (five). Deligne categories, Wick rotation in c, resonance/MC4 completion, Borel resummation, and spectral curves. Each fails for a precise reason: Deligne categories interpolate in a categorical variable, not a spectral one; Wick rotation changes the shadow coefficients but not the sewing lift $S_{\mathcal{A}}(u)$, which is c-independent; MC4 completion solves a homological algebra problem, not a number-theoretic one; Borel resummation requires factorial divergence that the shadow series does not exhibit; and spectral curves encode genus-0 OPE data, not the automorphic spectrum.

Genuine mathematics, no bridge (two). The analytic continuation of the shadow tower and the analytic Langlands programme produce real mathematical content but do not bridge the real-to-complex gap. The shadow tower continuation produces quadratic number fields (the Galois structure of the shadow extension K_L/F, with discriminant $\mathrm{disc}(Q_L) = -320\,c^2/(5c{+}22)$ for Virasoro). The analytic Langlands provides the correct framework for the critical-level oper/shadow interpolation. Both operate on the function-field (geometric) side; the number-field (arithmetic) descent remains the deepest unsolved problem.

The most compatible approach (one). The Connes–Marcolli non-commutative geometry programme has the closest structural compatibility with the bar construction: the Bost–Connes system has a Hecke algebra of symmetries compatible with Verdier–Hecke commutation, and the Weil positivity criterion is a trace-class condition on a specific operator. At the self-dual point $c = 13$, Verdier self-duality provides an involution on the factorisation coalgebra that could in principle feed into the non-commutative trace formula. In practice, the arithmetic descent (the bar complex is defined over $\mathbf{C}$, the Bost–Connes system over $\mathbf{Q}$) blocks the construction.

36.3 What the gravitational bar complex provides

The bar complex of the Virasoro algebra at central charge c provides five kinds of arithmetic data, all proved.

1. The exact genus expansion $F_g = (c/2)\,\lambda_g^{\mathrm{FP}}$ at all genera.
2. The infinite shadow tower with growth rate $\rho(c) = \sqrt{(36 + 2\Delta)/(c^2)}$ and discriminant $\Delta = 80/(5c{+}22)$.
3. The four-class depth assignment (class M for all $c \neq 0$).
4. The quadratic number field $K_L = \mathbf{Q}(\sqrt{\mathrm{disc}(Q_L)})$ at each rational c, with class numbers ranging from $h = 1$ (models $(9, 10)$ and $(24, 25)$) to $h = 3008$ (model $(28, 29)$).
5. Route C prime-locality: the MC recursion propagates Hecke equivariance from low-arity character data to all arities.

None of these constrains the zeros of the Riemann zeta function or any standard L-function. The constraints are algebraic (on OPE coefficients and their MC recursions); the zero locations are analytic (determined by the Euler product and the functional equation, not by the OPE).

36.4 The gravitational dichotomy and the spectral divide

The gravity dichotomy (critical string $c = 26$ versus non-critical $c \neq 26$) is a statement about the *vanishing* of a specific class: $\kappa_{\mathrm{eff}} = \kappa(\mathrm{matter}) + \kappa(\mathrm{ghost}) = 0$ if and only if $c = 26$. The self-dual point $c = 13$ is where the Koszul asymmetry $\delta_\kappa = \kappa - \kappa' = 0$ vanishes. These are different conditions at different central charges (AP29).

The gravity dichotomy operates on the *algebraic* axis: whether the Clifford algebra $\mathrm{Cliff}(V, \lambda)$ with $\lambda = \kappa(\mathcal{B})\,\omega_g$ degenerates to an exterior algebra ($\lambda = 0$ at $c = 26$) or remains a matrix algebra ($\lambda \neq 0$ at $c \neq 26$). This is a statement about the transgression algebra, not about the scattering matrix. The spectral divide (real versus complex spectral parameters) operates on the *analytic* axis. The two axes are orthogonal, and neither the gravitational dichotomy nor the Koszul self-duality provides a bridge between them.

36.5 The Beilinson verdict for quantum gravity

The following claims about the gravitational theory are dismissed.

1. "3d quantum gravity from Koszul duality constrains zeta zeros." *False.* The Koszul duality of the Virasoro algebra determines the complementary algebra Vir_{26-c} and the complementarity identity $\kappa + \kappa' = 13$. These are algebraic constraints on OPE invariants, not analytic constraints on L-function zeros.

2. "The gravitational genus expansion encodes arithmetic data beyond the character." *Partially true, partially false.* The genus expansion $F_g = \kappa\, \lambda_g^{\mathrm{FP}}$ is tautological (it lives in the tautological ring of $\overline{\mathcal{M}}_g$). The shadow tower provides additional algebraic data (the quartic resonance class, the discriminant, the depth classification). But none of this data accesses the primary spectrum or the partition function, which are the objects that carry arithmetic content through the Rankin–Selberg integral.

3. "The gravitational partition function at $c = 26$ has a distinguished relationship to the Riemann zeta function." *False.* The partition function of the $c = 26$ Virasoro is $\eta(\tau)^{-24} = \Delta(\tau)^{-1}$, the inverse of the Ramanujan discriminant form. The Rankin–Selberg L-function of Δ is $L(s, \Delta \times \Delta) = \zeta(s-11)\, L(s, \mathrm{Sym}^2 \Delta)$, which has critical line $\mathrm{Re}(s) = 23/2$, not $\mathrm{Re}(s) = 1/2$. The zeta function $\zeta(s-11)$ appearing here is shifted by 11 (the weight of Δ minus 1). The relationship to the Riemann zeta is classical (Hecke theory), not a consequence of the bar construction.

36.6 What survives for the gravitational theory

The bar complex of the Virasoro algebra at central charge c is the algebraic engine of 3d quantum gravity on $\mathbf{C}_z \times \mathbf{R}_t$. It determines: the Swiss-cheese algebra structure (bar differential = chiral product in $\mathbf{C}$, bar coproduct = topological cutting in $\mathbf{R}$), the curved genus tower ($d^2 = \kappa\, \omega_g$ at genus $g \geq 1$), the infinite shadow Postnikov tower (all arities, all genera, proved by Theorem **??**), the complementarity decomposition (Theorem C), and the completed platonic datum $\Pi_X^{\mathrm{oc}}(\mathcal{A})$ packaging open/closed/modular data.

These are genuine results about the algebraic structure of the gravitational theory. They do not imply, and should not be claimed to imply, any result about the analytic number theory of L-functions. The bar complex is the algebraic engine. The primary spectrum is the analytic output. The gap between them is the gap between algebra and analysis, which no known construction bridges.